Chapter 7

Biological platforms for the production of antimicrobial peptides: Bacterial, yeast, and cell-free systems

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Antimicrobial peptides (AMPs) play critical roles in the innate immune system (Ganz, 2003). They are short chains of amino acids, typically consisting of less than 100 residues, are primarily hydrophobic, and have a net positive charge due to an overabundance of basic residues (Bahar & Ren, 2013). AMPs exhibit a wide range of antimicrobial activities, with the ability to kill a broad range of microorganisms, including bacteria, viruses, fungi, and parasites. Furthermore, due to their distinctive mode of action, AMPs are minimally toxic to host cells. This makes them desirable candidates for developing new antibiotics to combat drug-resistant infections due to their selective toxicity (Hancock & Sahl, 2006).

Before AMPs can be examined in clinical trials, several fundamental questions, including mechanisms, efficacy, and safety, must be resolved. To answer these concerns, considerable functional and structural research is necessary, the advancement of which is partly dependent on the availability of obtaining pure peptides. The labor-intensive and time-consuming procedure of isolating peptides from natural sources is inefficient for obtaining large amounts of peptides, and chemical synthesis is difficult and costly (Andersson et al., 2000). Therefore, recombinant DNA technology permits inexpensive protein manufacturing and has enabled the mass production of AMPs (Ingham & Moore, 2007).

This chapter will focus on bacterial, yeast, and cell-free systems as biological platforms for the production of AMPs. We will discuss the benefits and limitations of each platform, as well as the obstacles that must be overcome to increase their efficiency and decrease production costs. Furthermore, we will highlight the most promising AMPs produced using these platforms and their potential applications in the development of novel antimicrobial therapies.

1 Bacterial hosts for antimicrobial peptide production

E. coli has been the most popular host among the various systems for producing recombinant proteins (Li & Chen, 2008). This bacterium has proven to be the most cost-effective method of recombinant protein production due to its rapid growth, wide availability of commercial expression vectors, well-established DNA manipulation protocols, and extensive genetic, biochemical, and physiological knowledge. It remains the preferred host for AMP production, with the highest yields reported in Table 7.1. In some other instances, other peptides are used, for example, *Bacillus subtilis*, a Gram-positive bacterium, was used instead of *E. coli* to express the hybrid peptide cecropin AD, composed of cecropin A and D. Furthermore, it has also been reported that *Propionibacterium freudenreichii* serves as a host for heterologous expression of AMPs. By developing a system for DNA transfer between *E. coli* and *P. freudenreichii*, it was possible to express bacteriocins successfully.

Some obstacles must be overcome to achieve efficient AMP production in bacteria. The first step is to prevent the natural activity of the AMP in order to avoid its lethality in the host strain. The second is to overcome the instability of the AMP due to its chemical properties and size; proteases synthesized by bacterial strains target AMPs. In light of this, two strategies for biosynthesizing AMPs are commonly employed: masking the AMP and direct expression.

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TABLE 7.1 Antimicrobial peptides expressed in several hosts.

Name	Source	Host	Direct Expression?	Masked?	Expression	Carrier protein	Cleavage site	Yield (mg/L)	References
ABP-CM4	*Bombyx mori*	*Pichia pastoris*	Y	N	Induced	–	–	ND	Zhang et al. (2006)
Adenoregulin	Frog *Phyllomedusa bicolor*	*Escherichia coli*	N	Y	Induced	Thioredoxin	Enterokinase	ND	Cao et al. (2005)
Adenoregulin	Frog *P. bicolor*	*E. coli*	N	Y	Induced	Thioredoxin	CNBr	ND	Zhou, Cao, Luo, et al. (2005)
Antifungal Protein (AFP)	*Aspergillus giganteus*	*Pichia pastoris*	Y	N	Induced	–	–	2.5 (MP)	López-García et al. (2010)
BmK AS	Scorpion *Buthus martensii* Karsch	*E. coli*	N	Y	Induced	His-tag	–	4.2 (MP)	Shao et al. (2008)
Brevinin-2R	Frog *Rana ridibunda*	*E. coli*	N	Y	Induced	Thioredoxin	Factor Xa	ND	Mehrnejad et al. (2008)
Buforin II/ Magainin	Frog *Bufo bufo gargarizans*/Frog *Xenopus laevis*	*E. coli*	N	Y	Induced	MMIS	CNBr	107 (MP)	Lee et al. (1998a, 1998b)
Buforin IIb	Frog *B. bufo gargarizans*	*E. coli*	N	Y	Induced	Thioredoxin	Hydroxylamine hydrochloride	3.1 (MP)	Wang et al. (2011)
Cathelicidin CAP18LL-37	Human	*Pichia pastoris*	Y	N	Constitutive	–	–	ND	Hong et al. (2007)
Cathelicidin CAP18LL-37	Human	*Pichia pastoris*	Y	N	Induced	–	–	ND	Kim et al. (2009)
Cecropin	*Musca domestica*	*E. coli*	N	Y	Induced	GST	Thrombin	ND	Liang et al. (2006)
Cecropin	*M. domestica*	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	48 (FP) 11.2 (MP)	Xu, Jin, Yu, Ji, et al. (2007)
Cecropin A (1–8)/ Magainin 2 (1–12)	Moth *Hyalophora cecropia*/Frog *X. laevis*	*E. coli*	N	Y	Induced	Ubiquitin	UCH	36 (FP)	Xu, Jin, Yu, Ren, et al. (2007)
Cecropin A (Mdcec)	*M. domestica*	*Pichia pastoris*	Y	N	Induced	–	–	20 (MP)	Jin et al. (2006)
Cecropin AD	Insect *H. cecropia*	*E. coli*	N	Y	Induced	SUMO	Sumo protease	30.6 (MP)	Chen et al. (2009)

Cecropin AD	Chimeric antimicrobial peptide obtained from cecropins	*Bacillus subtilis*	N	Y	Induced	SUMO	Sumo protease	30.6 (MP)	Chen et al. (2009)
Cecropin CM-4	Silkworm *B. mori*	*E. coli*	N	Y	Induced	SUMO	Hydroxylamine hydrochloride	252 (FP) 48 (MP)	Li et al. (2011)
Cecropin CM-4	Silkworm *B. mori*	*E. coli*	N	Y	Induced	SUMO	Sumo protease	112 (FP) 24 (MP)	Li et al. (2009)
Cecropin CM-4	Silkworm *B. mori*	*E. coli*	N	Y	Induced	Thioredoxin	Formic acid	NA (FP) 1.2 (MP)	Li, Zhang, et al. (2007)
Cecropin CM-4	Silkworm *B. mori*	*E. coli*	N	Y	Induced	Thioredoxin	Hydroxylamine hydrochloride	170 (FP) 68 (MP)	Zhou, Zhao, et al. (2009)
Cecropin CM-4	Silkworm *B. mori*	*E. coli*	N	Y	Induced	Thioredoxin	Hydroxylamine hydrochloride	1.4 (MP)	Zhou, Lin, et al. (2009)
Cecropin XJ	*B. mori*	*E. coli*	N	Y	Induced	Thioredoxin		10	Xia et al. (2013)
Cecropina pxCECA1	Moth *Plutella xylostella*	*E. coli*	N	Y	Induced	Intein	Self-cleavage	12.3 (MP)	Wang et al. (2012)
Ch-Penaeidin (rCHP)	*Fenneropenaeus chinensis*	*Pichia pastoris*	Y	N	Induced	–	–	180 (MP)	Li et al. (2005)
Crustin	Crab *Scylla paramamosain*	*E. coli*	N	Y	Induced	His-tag	–	ND	Imjongjirak et al. (2009)
Crustin	Shrimp *F. chinensis*	*E. coli*	N	Y	Induced	His-tag	Thrombin	ND	Sun et al. (2010)
Defensin	Oyster *Crassostrea gigas*	*E. coli*	N	Y	Induced	His-tag	CNBr	1 (MP)	Gueguen et al. (2006)
Defensin (Pdc1)	Corn inbred line CO387	*Pichia pastoris*	Y	N	Constitutive	–	–	ND	Kant et al. (2009)
Defensin (Psd1)	*Pisum sativum*	*Pichia pastoris*	Y	N	Induced	–	–	63 (MP)	Cabral et al. (2003)
Defensin (VrD1)	Mugbean	*Pichia pastoris*	Y	N	Induced	–	–	ND	Chen et al. (2004)
Defensin SPE10	*Pachyrrhizus erosus* seeds	*Pichia pastoris*	Y	N	Induced	–	–	7.0 (MP)	Song et al. (2005)

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TABLE 7.1 Antimicrobial peptides expressed in several hosts—cont'd

Name	Source	Host	Direct Expression?	Masked?	Expression	Carrier protein	Cleavage site	Yield (mg/L)	References
Dermcidin-1 L	Human	*E. coli*	N	Y	Induced	Thioredoxin	Factor Xa	ND	Lai et al. (2005)
Enterocin L50 (EntL50A)	*Enterococcus faecium* L50	*Pichia pastoris*	Y	N	Induced	–	–	228.5 (MP)	Basanta et al. (2010)
Enterocin P	*E. faecium*	*Pichia pastoris*	Y	N	Induced	–	–	22.8 (MP)	Gutiérrez et al. (2005)
Gallinacin 9	Chicken	*E. coli*	N	Y	Induced	GST	–	39.4 (FP) 210 (MP)	Ma et al. (2008)
Hepcidin	Flounder *Paralichthys olivaceus*	*E. coli*	N	Y	Induced	His-tag	TEV protease	89.1 (FP)	Srinivasulu et al. (2008)
Hepcidin	Human	*E. coli*	N	Y	Induced	Thioredoxin	Thrombin	ND	Gagliardo et al. (2008)
Hinnavin II	Butterfly *Artogeia rapae*	*E. coli*	N	Y	Induced		Enterokinase	15 (FP)	Kang et al. (2008)
Hinnavin II/α-melanocyte-stimulating hormone (Hybrid)	Butterfly *Pieris rapae*	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	260 (MP)	Bang et al. (2010)
Histonin/[KLLK]2RLKR)	Frog *B. bufo gargarizans*	*E. coli*	N	Y	Induced	PurF protein	Furin protease	167 (MP)	Kim et al. (2008)
IFN SasalFN-alpha1	*Salmo salar*	Rice and potato	Y	N	Constitutive	–	–	–	Fukuzawa et al. (2010)
Indolicidin	–	*E. coli*	N	Y	Induced	Thioredoxin	CNBr	0.15 (MP)	Morin et al. (2006)
Lactoferrampin fused with	Bovine	*Pichia pastoris*	Y	N	Induced	–	–	ND	Tang et al. (2012)
Lactoferricin	Bovine	*E. coli*	N	Y	Induced	MMIS	CNBr	60 (MP)	Kim et al. (2006)
Lactoferricin	Bovine	*E. coli*	N	Y	Induced	GST	Thrombin	ND	Luo et al. (2007)
Lactoferricin W10	Bovine	*E. coli*	N	Y	Induced	GST	Enterokinase	20 (FP) 300 (MP)	Feng et al. (2012)

Lactoferricin/ AMP-HP	Bovine/HP – *Helicobacter pylori*	*E. coli*	N	Y	Induced	Thioredoxin	Formic acid	11.3 (MP)	Tian et al. (2009)
Lactoferricin/ Thanatin (hybrid)	Bovine (LfcinB)	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	0.5 (FP)	Feng et al. (2010)
Latarcin 2a	Spider *Lachesana tarabaevi*	*E. coli*	N	Y	Induced	Thioredoxin	CNBr	3.2 (MP)	Shlyapnikov et al. (2008)
LFT33	Derived from LfcinB and thanatin	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	0.5 (MP)	Feng et al. (2012)
LL-37	Human	*E. coli*	N	Y	Induced	GST	Factor Xa	8 (FP) 0.3 (MP)	Moon et al. (2006)
LL-37	Human	*E. coli*	N	Y	Induced	Thioredoxin–SUMO dual-tag		2.4	Yifeng Li (2013)
Mi AMP1	Plant *Macadamia integrifolia*	*E. coli*	N	Y	Induced	–	–	2.5 (MP)	Harrison et al. (1999)
OG2	Frog *Odorrana grahami*	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	50 (FP)	Xie et al. (2013a)
Palustrin-2CE	Frog *Rana chensinensis*	*E. coli*	N	Y	Induced	GST	Enterokinase	ND	Sun et al. (2012)
Pediocin PA-1	*Pediococcus acidilactici*	*Saccharomyces cerevisiae*	Y	N	Constitutive	–	–	ND	Schoeman et al. (1999)
Perinerin	Clamworm *Perinereis aibuhitensis* Grube	*E. coli*	N	Y	Induced	Thioredoxin	Factor Xa	60 (FP) 1.2 (MP)	Zhou et al. (2007)
Pg-AMP1	Guava *Psidium guajava*	*E. coli*	N	Y	Induced	His-tag	–	1 (MP)	Tavares et al. (2012)
Plantaricin 423	*Lactobacillus plantarum* 423	*S. cerevisiae*	Y	N	Constitutive	–	–	ND	Van Reenen et al. (2003)
Plectasin	Fungus *Pseudoplectania nigrella*	*E. coli*	N	Y	Induced	Thioredoxin	Factor Xa	92 (FP) 3.5 (MP)	Jing et al. (2010)
Plectasin	*Pseudoplectania nigrella*	*Pichia pastoris*	Y	N	Induced	–	–	748.63 (MP)	Zhang et al. (2011)
Pleurodicin	*Pleuronectes americanus*	*Pichia pastoris*	Y	N	Induced	–	–	0.01 (MP)	Burrowes et al. (2005)

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TABLE 7.1 Antimicrobial peptides expressed in several hosts—cont'd

Name	Source	Host	Direct Expression?	Masked?	Expression	Carrier protein	Cleavage site	Yield (mg/L)	References
Porcine interferon-a (pIFN-a)	Porcine	*Pichia pastoris*	Y	N	Constitutive	–	–	ND	Yu et al. (2010)
Pp AMP	Pine *Pinus pinaster*	*E. coli*	N	Y	Induced	His-tag	–	6 (FP)	Canales et al. (2011)
Puroindoline A	Wheat seed	*E. coli*	N	Y	Induced	GST	Thrombin	2.6 (FP) 1.8 (MP)	Capparelli et al. (2007)
Sarcotoxin IA	Fly *Sarcophaga peregrina*	*E. coli*	N	Y	Induced	GFP	CNBr	0.5 (MP)	Skosyrev et al. (2003)
Scygonadin	Crab *Scylla serrata*	*E. coli*	N	Y	Induced	CKS	3C Protease	89.1 (FP) 10.6 (MP)	Peng et al. (2010)
Scygonadin	*Cherax quadricarinatus*	*Pichia pastoris*	Y	N	Induced	–	–	89.2 (MP)	Peng et al. (2012)
SMAP-29	Sheep	*E. coli*	N	Y	Induced	InteinCBD	Self-cleavage	ND	Morassutti et al. (2005)
SpStrongylocins 1 and 2	Urchin *Strongylocentrotus purpuratus*	*E. coli*	N	Y	Induced	His-tag	Enterokinase	ND	Li et al. (2008)
Thanatin	Insect *Podisus maculiventris*	*E. coli*	N	Y	Induced	Thioredoxin	Thrombin	13.2 (MP)	Wu et al. (2008)
Viscotoxin	Mistletoe *Viscum album*	*E. coli*	N	Y	Induced	Thioredoxin	TEV protease	31 (FP) 5.2 (MP)	Bogomolovas et al. (2009)
Vv-AMP1	Plant *Vitis vinifera*	*E. coli*	N	Y	Induced	GST	Thrombin	2.6 (FP) 1.8 (MP)	De Beer and Vivier (2008)
α-Defensin (HD5)	Human	*Pichia pastoris*	Y	N	Induced	–	–	165 (MP)	Wang et al. (2009)
α-Sarcin	*Aspergillus giganteus*	*Pichia pastoris*	Y	N	Induced	–	–	ND	Martínez-Ruiz et al. (1998)

β-Defensin (msBD-1)	Sheep	*Pichia pastoris*	Y	N	Induced	–	–	35 (MP)	Zhao and Cao (2012)
β-Defensin 12	Bovine	*E. coli*	N	Y	Induced	Thioredoxin	–	0.05 (FP)	Wu et al. (2011)
β-Defensin 2	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	346 (FP) 100 (MP)	Xu, Peng, et al. (2006)
β-Defensin 2	Human	*E. coli*	N	Y	Induced	Thioredoxin	CNBr	900 (FP) 630 (MP)	Zhong et al. (2006)
β-Defensin 2 (hPAB-B)	Human	*Pichia pastoris*	Y	N	Induced	–	–	241 (MP)	Chen, Wang, et al. (2011)
β-Defensin 26	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	1.380 (FP) 524 (MP)	Hernandez-Garcia et al. (2010)
β-Defensin 27	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	1.290 (FP) 464 (MP)	Huang et al. (2009)
β-Defensin 3	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	900 (FP) 210 (MP)	(Huang et al. (2006)
β-Defensin 4	Human	*E. coli*	N	Y	Induced	SUMO	Sumo protease	637 (FP) 166 (MP)	Li, Zhang, et al. (2010)
β-Defensin 4	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	2.680 (FP) 689 (MP)	Xu, Zhong, et al. (2006)
β-Defensin 5	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	1.490 (FP) 566 (MP)	Huang et al. (2008)

Continued

TABLE 7.1 Antimicrobial peptides expressed in several hosts—cont'd

Name	Source	Host	Direct Expression?	Masked?	Expression	Carrier protein	Cleavage site	Yield (mg/L)	References
β-Defensin 6	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	1.570 (FP) 549 (MP)	Harrison et al. (1999)
β-Defensin 6	Human	*E. coli*	N	Y	Induced	Thioredoxin	Enterokinase	1.690 (FP) 380 (MP)	Wang et al. (2010)
β-Defensin-1	Human	*S. cerevisiae*	Y	N	Constitutive	–	–	0.055 (MP)	Cipakova and Hostinova (2005)
θ-Defensin-1	*Rhesus macaques*	*E. coli*	N	Y	Induced	Gyrase intein	GSH	5 (FP) 0.01 (MP)	Gould et al. (2012)

1.1 Masking the AMP: Using protein fusion strategies

The antimicrobial nature of the peptides makes them potentially harmful to the producing host, posing a significant challenge in their biological production. Furthermore, peptides are particularly vulnerable to proteolytic degradation because of their small size and high cationic charge. These two obstacles can be circumvented by fusing the required peptide with a carrier protein. Enzymatic or chemical cleavage at the appropriate point near the carrier-peptide junction can release the target peptide from the fusion. The fusion structure mirrors the structure of the peptide precursors during their natural biosynthesis by using a carrier protein that behaves similarly to the peptide's native prosegment and protects the peptide from proteases and the host from the hazardous peptide (Vassilevski et al., 2008). There are two common carriers: *those that improve solubility* and *those that promote aggregation*. In addition to these two major groups, self-cleavable proteins and secretion signals are unique carriers that allow auto-tag removal and, correspondingly, secretory expression (Fig. 7.1).

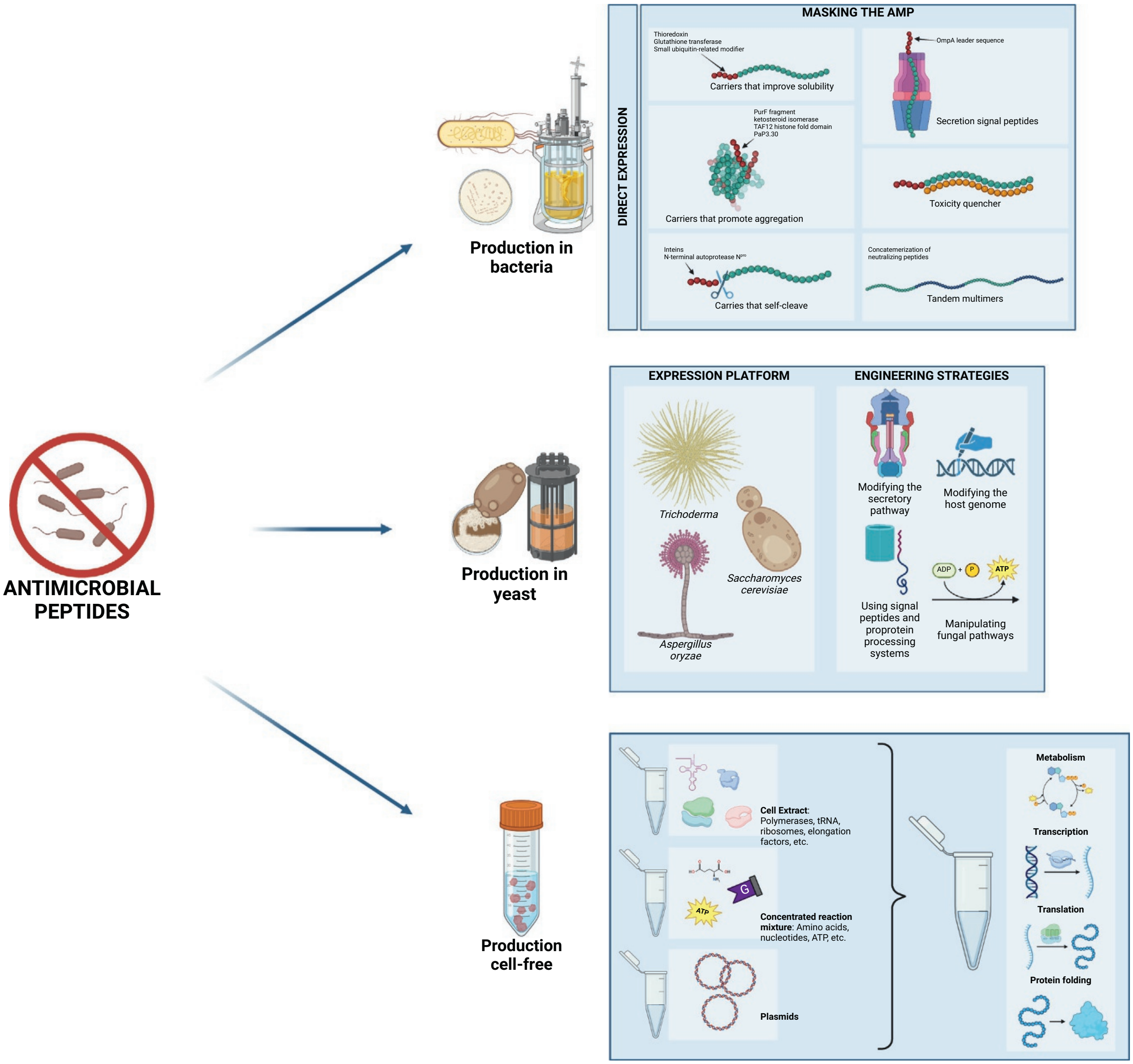

FIG. 7.1 Comparative strategies for antimicrobial peptide production in biological systems: from bacteria to cell-free approaches. This figure illustrates the diverse strategies for antimicrobial peptide production across various biological systems, including bacteria (A), yeast (B), and cell-free systems (F).

1.1.1 *Carrier proteins*

Carriers that improve solubility

Thioredoxin and glutathione transferase (GST) are the most commonly used carrier proteins for AMP fusion expression because of their robust ability to promote the soluble expression of recombinant proteins/peptides in the cytoplasm of *E. coli* (LaVallie et al., 1993, 2000; Sheibani, 1999). Even though both proteins are equally effective in increasing the solubility of fusion proteins, thioredoxin is preferred for peptide synthesis due to its small size (11.8 kDa), which allows the target to constitute a relatively large fraction of the fusion. Among the 13 potential carrier proteins studied, thioredoxin fusion had the highest absolute yield of the fused peptide, while it did not have the highest expression level overall (Bogomolovas et al., 2009). Despite their widespread popularity, GST fusions have been reported to be particularly vulnerable to proteolytic degradation, resulting in inefficient or failed peptide synthesis (Chen et al., 2008; Si et al., 2007; Skosyrev et al., 2003; Zhou, Cao, Wang, et al., 2005). Small ubiquitin-related modifier (SUMO) is a relatively recent fusion carrier that enhances protein folding and solubility (Butt et al., 2005; Malakhov et al., 2004). Similar to thioredoxin, its modest size (11.2 kDa) allows for a relatively high peptide-to-carrier ratio, which is advantageous for peptide yield. In addition, a specialized SUMO protease promotes the effective release of the passenger protein or peptide, which is the significant advantage of this approach. In addition to serving as a fusion transporter, GST also functions as an affinity tag to purify the fusion on immobilized glutathione. In contrast, thioredoxin and SUMO require extra affinity tags for the initial purification of the fusion proteins. The His-tag is the most commonly used affinity tag for this purpose (Ingham & Moore, 2007).

Carriers that promote aggregation

Unlike solubility-enhancing carriers, several proteins are selected as carriers due to their high tendency to form inclusion bodies, which are insoluble aggregates of protein that accumulate within cells, often as a result of protein misfolding or overexpression, visible under a microscope as distinct granules. The PurF fragment, ketosteroid isomerase, TAF12 histone fold domain, and PaP3.30 are members of this group (Lee et al., 2000; Rao et al., 2004; Zorko & Jerala, 2010). It is hypothesized that insoluble expression is more successful at masking the harmful effects of peptides and protecting them from degradation than soluble fusion. Also, insoluble expression enables the fusion to be purified rapidly through cell lysate centrifugation. Fusion proteins are often the most insoluble elements; therefore, additional purification may not be required beyond simple washing. A small percentage of insoluble fusions contain His-tag, allowing purification by immobilized metal-affinity chromatography (IMAC) under denaturing conditions. Cysteine-free AMPs are compatible with carriers prone to aggregation, as refolding is usually unnecessary to restore activity.

Carries that self-cleave

Inteins are auto-processing domains found in organisms from all domains of life. Due to their potential to extricate themselves from a precursor protein and rejoin flanking areas (Kane et al., 1990; Perler et al., 1994), they are regarded as the protein counterpart of introns. After a few modifications, inteins can be utilized as self-cleavage components that undergo selective N- or C-terminal cleavage induced by thiol reagents, pH, and temperature changes (Xu & Evans, 2003). Most of the time, soluble intein fusions can be obtained when cultures are grown at low temperatures (i.e., 20°C–30°C) Xie et al., 2013a, 2013b. New England Biolabs offers two commercial systems: Sce VMA intein and Ssp dnaB-derived mini-intein https://www.neb.com/en/-/media/nebus/files/pdf-faq/impact-faqs.pdf?rev=29d1d5b1b27f4312a0a5dedba9bca372&hash=C3B24E6D92D52631D04DA4DBEC030BA5. The inteins in both systems are fused to a CBD affinity tag (5.6 kDa). Cleavage can produce target proteins with native N-termini, essential for synthesizing AMPs. Mini-intein is ideal for producing small peptides because it allows for a greater stoichiometric ratio of the target. Cheng et al. purified the recombinant AMP CM4 using a self-cleavable fusion tag derived from the N-terminal autoprotease N^{pro} (Cheng et al., 2010). The fusion protein was deposited as inclusion bodies, and the target peptide was released through Npro-mediated self-cleavage. The target peptide can be purified using relatively straightforward procedures thanks to the intein and N^{pro} systems, which do not require exogenous proteases or other substances. Additionally, even at low concentrations, uncontrolled auto-cleavage of intein fusions could be a problem because the released AMPs can be lethal when they come into direct intracellular contact with their targets (Lee et al., 2000). The N^{pro} system has several disadvantages, including extensive dilution, a lengthy incubation period, and incomplete cleavage (Li, 2011b).

Secretion signal peptides

In several instances, signal peptides (alone or in conjunction with carrier proteins) have been added to the N-terminus of the target AMP to facilitate periplasmic secretion. The periplasm is more favorable to disulfide bond production than the cytoplasm and has been widely employed to promote protein folding. Lee et al. achieved periplasmic production of soluble hepcidin I, a cysteine-rich AMP from olive flounder, by utilizing the OmpA leader sequence with enhanced hydrophilicity as a signal (Lee et al., 2008).

Coexpression with a toxicity quencher

The coexpression system created by Jang et al. can be seen as an additional type of fusion (Jang et al., 2009). The target peptide was coexpressed in their design with an anionic protein prone to aggregation. The coexpression partner complexed with the target peptide and bolstered the creation of inclusion bodies. Like the common fusion carrier, the coexpression partner protected the peptides from proteolytic breakdown and sheltered them from their fatal consequences. This innovative design eliminates the cleavage step required for peptide retrieval, which is a benefit over the conventional fusion method. The dissociated peptide can be easily extracted by cation-exchange chromatography following the solubilization of the complex in urea. One liter of bacterial culture yielded 100 mg of peptide for each of the three tested peptides.

Tandem multimers

Several AMPs have been expressed as tandem multimers rather than monomers (Kim et al., 2008; Lee et al., 1998a, 1998b; Morin et al., 2006; Rao et al., 2005; Sun et al., 2004; Tian et al., 2009; Wang & Cai, 2007; Zhong et al., 2006). Initially, buforin II was fused with a comparable-sized acidic peptide, and the resulting fusion was expressed as tandem repeats (Lee et al., 1998a, 1998b). By neutralizing its positive charge, the acidic peptide effectively concealed buforin II's toxicity. Subsequent experiments revealed that simple concatemerization typically results in poorly expressed proteins, particularly when the copy number exceeds two, and that charge-neutralizing peptides and proteins are essential for expression (Morin et al., 2006; Sun et al., 2004). Using the same design as buforin II, two additional peptides, lactoferricin, and the hybrid cecropin-melittin, were expressed (Kim et al., 2006; Wang & Cai, 2007). Another explored route to produce AMPs tandemly is using a linked peptide multimer fused to a carrier protein (such as thioredoxin) (Metlitskaia et al., 2004; Morin et al., 2006; Tian et al., 2009; Zhou, Zhao, et al., 2009). In both instances, the peptide junction featured designed cleavage sites that permitted monomer release when treated with the appropriate chemical reagent. Because nearly all multimeric fusions generated inclusion bodies, chemical cleavage was utilized significantly more frequently than enzymatic cleavage.

Peptide cleavage

Except for self-cleavage carriers (intein and N^{pro} fusions), which undergo auto-cleavage upon induction, the release of the target peptides from fusion proteins requires exogenous chemicals or proteases. Insoluble fusions are mostly cleaved by chemicals that can be directly applied to the purified inclusion bodies. Either type of reagent can break apart soluble fusions. Piers et al. first reported that enzymatic cleavage was less effective than chemical methods (Piers et al., 1993). This inefficiency may be attributable to unfavorable surrounding residues or the peptide segment's tendency to aggregate, making the cleavage sites less accessible (Li, Li, et al., 2007; Zhou et al., 2007). Dialysis with 8 M urea before cleavage can enhance the efficiency of enzymatic cleavage in some instances (Qian et al., 2000).

Cyanogen bromide (CNBr), formic acid, and hydroxylamine are the three most commonly used chemicals for fusion cleavage (Duvick et al., 1992; Ingham & Moore, 2007; Zorko & Jerala, 2010). CNBr cleaves peptide bonds on the carboxyl side of methionine to produce homoserine; however, exposure to formic acid may result in side chain modification. Formic acid cleaves between Asp and Pro and can cause the esterification of serine and threonine residues. The Asn-Gly peptide bond is broken by hydroxylamine. Nevertheless, Asn and Gln can be converted to their hydroxamate forms. Although CNBr has been used more frequently than the other two, its use is limited because many AMPs contain methionine in their sequence (Zorko & Jerala, 2010). Additionally, subsequent purification may be more challenging and ineffective if methionines are present in the carrier or linker region. Both formic acid and hydroxylamine recognize residue pairs and are less likely to cleave at unwanted sites (Li, 2011a; Li et al., 2006). Chemical cleavage's disadvantages include toxicity and side chain modifications (Li, 2009).

Several proteases have been utilized to promote proteolytic cleavages. Enterokinase (AspAspAspAspLys↓), Factor Xa (IleGluGlyArg↓), and Thrombin (LeuValProArg↓GlySer) are commonly used proteases. However, their sensitivity to pH and chaotropic agents has been described (Havarstein et al., 1995). Unlike other proteases, the SUMO protease recognizes the tertiary structure of SUMO rather than short sequence motifs. Consequently, the enzyme never cleaves within the target protein, allowing for the production of a target protein with a native N-terminus. Moreover, the enzyme is exceptionally robust and retains most of its activity in 2 M urea. As a result, SUMO protease can cleave urea-solubilized and diluted insoluble fusions. The protease from the tobacco etch virus (TEV) (GluAsnLeuTyrPheGln [Gly/Ser]) is another enzyme that is not inhibited by urea at low concentrations (1 M). TEV protease, like SUMO protease, can generate a native N-terminus in many instances. A major benefit of the SUMO and TEV proteases is their resistance to urea, which can prevent protein aggregation and loosen compact structures, making cleavage easier. Another advantage is the relatively easy production of large amounts of SUMO and TEV proteases.

Generally, high-performance liquid chromatography (HPLC) is used to purify the released peptides. Since the pI values of AMPs are typically significantly higher than those of the carrier proteins, the liberated peptides can also be efficiently isolated using cation-exchange chromatography.

1.2 Direct expression

In contrast to fusion expression, a direct expression is an approach that does not use carrier molecules. In this method, a peptide is directly produced, meaning that the product contains no extra sequences other than a His-tag and a tag removal recognition site (the His-tag on the target peptide may or may not be removed before further use). In most cases, the production of directly expressed peptides causes inclusion bodies (Amparyup et al., 2008; Harrison et al., 1999; Imjongjirak et al., 2009; Peng et al., 2010; Shao et al., 2008; Srinivasulu et al., 2008; Supungul et al., 2008). Due to the size and aggregation propensity, this approach minimizes potential toxicity and stability difficulties.

AMPs have also been expressed using nontraditional yeasts such as *Pichia pastoris*. In response to diverse stress conditions, the expression of genes encoding AMPs can be induced in these organisms (Garvey, 2022). The most common approach is to employ induction by cytokines and heavy metals. Although this strategy is effective in most cases, it relies on the availability of these chemicals and usually results in low expression levels of AMPs. Gene knockout of the endogenous genes encoding for AMPs is an alternative method. This is typically challenging, however, due to functional redundancy between related genes and the resulting absence of desirable phenotypes (Varsaki et al., 2015). Therefore, there is a real need to develop novel tools for producing and engineering AMPs using fungal hosts. To this end, several tools for expressing AMPs in *Trichoderma reesei* and other filamentous fungi have been developed recently. For instance, it is possible to increase the production of endogenous AMPs by engineering the fungal host to increase the secretion of these molecules into the culture medium. This approach has been highly successful in some species but has not been successful in other fungal species. To overcome this problem, it has been proposed to engineer the microorganism to express an exogenous gene encoding an AMP that is secreted efficiently into the culture medium.

2 Exploring the use of fungi for antimicrobial peptide production

Fungi as hosts for AMP production have received special attention in the last decade because these microorganisms can produce both host-defense peptides and AMPs and are amenable to genetic engineering techniques (Lu et al., 2021). However, the use of fungi for this purpose has met with limited success to date, primarily due to the difficulty of producing large quantities of peptide in the fermentation medium. This is especially true for fungi that lack a secretion system for polypeptide export into the culture medium. In light of this challenge, researchers have sought to identify strategies for increasing peptide production in fungi (Arnau et al., 2020; Ward, 2012). Utilizing fungal species that produce proteins through the SREBP-1 (sterol regulatory element-binding protein 1) pathway, which is regulated by the sterol regulatory element-binding protein 1 (SREBP-1) pathway, has been one successful approach to addressing this problem (Burr et al., 2017; Lu et al., 2021). The expression of heterologous proteins in fungi can be enhanced by modifying the secretory pathway and/or the host genome, thereby increasing the organism's peptide production capacity (Demain & Vaishnav, 2009; Gerngross, 2004; Mattanovich et al., 2011; Sakekar et al., 2021). Several methods have been employed to improve the secretion of recombinant proteins in filamentous fungi, including the use of signal peptide (SP) and proprotein processing systems, the manipulation of fungal pathways that regulate protein secretion, and the engineering of the host genome to produce greater amounts of recombinant proteins (Anné et al., 2012; Gündüz Ergün et al., 2019; den Haan et al., 2021; Paloheimo et al., 2003).

It has been reported that engineering *Saccharomyces cerevisiae* to produce AMPs in its culture medium increases the yield of secreted AMPs (Deng et al., 2017; Parachin et al., 2012). This approach was first tested in the yeast *P. pastoris*, which secretes its polypeptides directly into the growth medium using a system of secretory vesicles known as the peroxisome pathway. This secretory pathway was enhanced by engineering the organism to express heterologous SPs, increasing the expression and secretion of AMPs (Kim et al., 2015; Pham et al., 2019; Yalçınkaya, 2017). However, this mode of production is not widely used because it is technically difficult and expensive and because it is limited to producing a small number of AMPs at a time. In addition, this approach does not address the issue of scale-up production, making it difficult to produce larger amounts of AMPs in an efficient manner.

A second approach to producing AMPs in *P. pastoris* has been to engineer the yeast to produce AMPs in the endoplasmic reticulum (ER) and then secrete them into the culture medium (Ahmad et al., 2014; Baghban et al., 2019; Li, Anumanthan, et al., 2007). This system relies on the secretion of a protein known as Npr1p, which is involved in the folding of newly synthesized polypeptides before they enter the lumen of the ER for secretion. Overexpression of Npr1p has been shown to induce the expression of several AMP genes, inducing an AMP phenotype in cells lacking a functional secretory pathway. This strategy has been found to be more effective for producing higher levels of AMPs because the engineered yeast can secrete them directly into the culture medium from their cells. Recent studies have shown that this approach has been useful for producing recombinant *P. pastoris* strains expressing AMPs such as defensins and cecropins (Deo et al., 2022; Guo et al., 2012; Sang et al., 2017). During growth, *P. pastoris* can form pseudohyphae that separate into single cells. This property enables the rapid production of large numbers of cells. However, it necessitates costly media and unique culture conditions (Shrivastava et al., 2023). Despite *S. cerevisiae*'s popularity as a host for many recombinant proteins, *P. pastoris* has emerged as the preferred host for AMP expression, with the FDA's approval of the first recombinant drug produced with this nonfermentative yeast in 2009 (Karbalaei et al., 2020). Because *P. pastoris* can be grown at a higher cell density, most AMPs produced in yeast have been expressed there. Comparing the expression of the defensin-encoding gene PDC1 in *E. coli* and *P. pastoris* revealed that the protein produced in yeast inhibits growth activity by a factor of 2. Despite the fact that most studies on recombinant AMP production using *P. pastoris* are still done in shake flasks, it is clear that *P. pastoris* is the preferred yeast for host expression (Karbalaei et al., 2020).

Recently, *Trichoderma* fungi have been used to heterologously express AMPs such as defensins, lantibiotics, and cecropins because they grow faster and express their genes more strongly than other expression systems (Liang et al., 2022). BMGlv2, a short-neck clam AMP (*Cerastoderma glaucum*), was successfully expressed in *Trichoderma reesei*. This expression system allowed for the low-cost and rapid production of large amounts of BMGlv2 without the need for costly equipment or specialized culture conditions (Liang et al., 2022). BMGlv2 has been shown to be active against a variety of Gram-positive and Gram-negative bacteria, as well as fungi, and it was discovered that when expressed, BMGlv2 was exposed to temperatures above 40°C or pH levels below 6, it retained its antibacterial activity. It is, therefore, of potential use in several therapeutic applications, including wound healing and infection treatment, as well as being an effective tool for food preservation. This high temperature and low pH stability are key factors that set BMGlv2 apart from other AMPs, and its discovery has furthered our understanding of the biotechnological applications of AMPs.

Yarrowia lipolytica is an industrially relevant yeast strain that can be used for the recombinant production of peptides. For instance, the yeast *Y. lipolytica* SWJ-1b was used to produce recombinant AMP from the Zhikong scallop (*C. farreri*) using the expression plasmid pINA1317. The AMP-encoding DNA fragment was inserted into the expression plasmid, and the transformants were screened using PCR. Crude AMP was extracted from yeast transformant 29a grown in YPD or PBB medium using centrifugation. SDS-PAGE was used to analyze the AMP after it was purified using Ni-NTA spin column chromatography. The antibacterial activity of purified AMP was assessed using minimum inhibitory concentration (MIC) assays against a variety of bacterial species. The mechanism of action of the AMP was investigated using scanning electron microscopy on bacterial protoplasts. The Bradford method was used to determine the protein content of purified AMP (Zhao et al., 2013). *Y. lipolytica* is also able to express many of the enzymes involved in lipid metabolism and produce various endogenous peptide hormones. For these reasons, *Y. lipolytica* has become an important model organism for studying protein secretion mechanisms in fungi.

Filamentous fungi are thought to be efficient and powerful cell factories for industrial-scale protein production. Many filamentous fungi are generally regarded as safe (GRAS). Various strategies, such as optimizing the fermentation process and obtaining beneficial mutants through random mutagenesis, have been used in the past to improve the production of various proteins by filamentous fungi. This chapter, however, will focus on summarizing these strategies rather than providing a comprehensive view of achieving efficient protein production. The protein secretion pathway in filamentous fungi consists of three major steps: polypeptide transfer from the ribosome to the ER, protein folding and modification in the ER, and transport of folded protein vesicles to the Golgi apparatus and extracellular environment. Protein folding and modification in the ER require the help of molecular chaperones and folding enzymes. Nascent peptides are glycosylated to ensure proper folding. A large number of proteins are involved in the formation, transportation, and fusion of vesicles, with the specific fusion of vesicles with target membranes being a critical process mediated by the soluble N-ethylmaleimide-sensitive factor-associated protein receptor (SNARE). To improve protein expression and secretion in filamentous fungi, genetic engineering strategies such as utilizing the target protein's transcription and/or codon, replacing the original

SP with a more efficient one, and fusing a heterologous protein to a naturally secreted one can be used. Protein secretion can be increased by regulating the unfolded protein response (UPR) and endoplasmic reticulum-associated degradation (ERAD). Furthermore, optimizing the intracellular transport process, creating a protease-deficient strain, regulating mycelium morphology, and regulating SREBP can all help to increase protein expression and secretion. Overall, filamentous fungi have a strong protein secretion pathway and are appealing protein expression and secretion cell factories. Genetic engineering advances have enabled the rapid tracking and discovery of novel determinants, resulting in increased protein expression and secretion. Finally, filamentous fungi have a high potential for protein production, and further efforts in genetic engineering will almost certainly lead to further advancements in this field (Karagiosis & Baker, 2012; Li et al., 2022; Lübeck & Lübeck, 2022; Wang, Zhong, and Xiao, 2020).

Other fungi have also been exploited in the production of AMPs. In a recent study, for example, researchers created and tested a recombinant hybrid AMP called Mag II-CB in the *Ascomycetes* fungus *Cordyceps militaris*. To accomplish this, a binary T-DNA expression vector containing the hybrid AMP gene Mag II-CB was created. This expression vector was then transformed into *Agrobacterium tumefaciens*, yielding 30–60 transformants per 1×10^5 protoplasts. Following that, the total protein content of both wild-type (WT) *C. militaris* and different clones of transformed *C. militaris* were analyzed using Western blotting. The Mag II-CB and CB protein bands were found to be approximately 10.8 and 6.2 kDa in size, respectively. Mag II-CB and CB purification yields were 3.86 mg and 2.95 mg/g, respectively. The antibacterial activity of the Mag II-CB recombinant hybrid AMP was tested in vitro and found to be significant against Gram-negative and Gram-positive bacteria. Furthermore, when tested in mice, Mag II-CB was discovered to have antibacterial properties and to protect mice from bacteria through both direct antibacterial activities and immunomodulatory functions. The study's findings show that treatment with a Mag II-CB hybrid AMP reduced damage to the intestinal mucosal barrier in mice by inhibiting the decrease in tight junction protein expression caused by *E. coli* infection. When *C. militaris* mycelium-producing Mag II-CB was directly fed to mice infected with *E. coli*, it was found to have antibacterial and immunomodulatory activities (Zhang et al., 2018).

Another example is the recombinant production of plectasin using *Aspergillus oryzae*. Plectasin is the first antimicrobial defensin peptide discovered in a fungus, *Pseudoplectania nigrella*, which is commonly found on the floor of northern European pine forests. The structure of plectasin was determined using NMR spectroscopy, which revealed an alpha-beta motif composed of an alpha-helix and two antiparallel beta-strands held together by three disulfide bonds. Plectasin killed *S. pneumoniae* in vitro at rates comparable with penicillin and vancomycin, and it did not show cross-resistance with other commonly used antibiotics. Furthermore, plectasin was not toxic to mouse fibroblasts or human erythrocytes, indicating bacterial selectivity over mammalian cells. Plectasin was found to be biologically active even after incubation in serum, and intact, active plectasin was recovered from mouse urine. It effectively cured systemic pneumococcal infection in two mouse models with no adverse effects on the mice. The discovery of plectasin has significant implications for the development of defensins as antiinfective therapeutics, as the main challenge in commercial development has been the technical difficulty in producing them. More research is needed to determine the exact mechanism of plectasin's antimicrobial activity. The researchers cultured *P. nigrella* mycelia, extracted RNA, and generated cDNA for cloning and analysis (Mygind et al., 2005).

2.1 Expression systems

Recent advances in synthetic biology have enabled the development of several yeast-based synthetic systems for AMP production, which can serve as valuable models for designing large-scale systems for producing these essential molecules. A recent example is an expression system created in *P. pastoris* by inserting a gene of interest, which was a fusion protein consisting of a secretion signal, human serum albumin (HSA), and the AMP apidaecin, using a serine recombinase. The fusion protein was inserted after a methanol-inducible promoter into *P. pastoris*, and the correct insertion was confirmed using colony PCR. After that, the *P. pastoris* strain was cultured and induced to express the fusion protein, which was purified and tested for antimicrobial activity against *E. coli*. In a bioreactor, three stages were involved in the production of apidaecin AMP: inoculum seed preparation, biomass accumulation, and protein production. The cells grew quickly in the glycerol-fed-batch phase and slowly in the methanol-fed-batch phase. With a titer of more than 700 mg/L, the HSA-apidaecin fusion protein was successfully released into the culture supernatant (Cao et al., 2018).

Additionally, these approaches have greatly contributed to our understanding of the underlying cellular mechanisms for efficient recombinant protein production. For example, a genome-scale metabolic model was used to predict gene targets for deletion or overexpression. Several factors, such as the efficiency of translocation to the ER lumen, affect the early steps of the secretion pathway. The level of protein production has to be optimized to avoid saturation of the secretory capacity of the host cells. Overloading of ER folding and secretion capacity can lead to the accumulation of mis- or unfolded proteins (Thak et al., 2020).

Another strategy for enhancing the expression of recombinant proteins is codon optimization. Codon usage varies between organisms, with subsets of codons, termed optimal codons, being efficiently translated and almost exclusively used in highly expressed genes. Codon optimization based on individual codon usage bias (ICU bias) has been the standard approach, but in recent years, codon-pair context bias (CC bias) has gained traction as an efficient strategy for synthetic gene design. Several factors, including the average GC content of the host genome, the secondary structure of mRNA, the codon adaptation index, and the tRNA pool, should be considered during codon optimization. Integration of expression constructs into the chromosomes of the host organism is preferable to plasmid-based expression. Integration of DNA into the host genome through nonhomologous recombination or untargeted ectopic integration may result in the development of unwanted side effects. To avoid these drawbacks, it is preferable to insert genetic constructs into gene loci whose deletion has no effect on yeast growth. To avoid the disadvantages of nontargeted integration and position effect, it is more practical to integrate genetic constructs for recombinant protein expression into a noncoding genomic position of yeast cells. However, integration can have side effects if it occurs through nonhomologous recombination or untargeted ectopic integration. To avoid these issues, it is preferred to insert genetic constructs into gene loci that do not affect yeast growth. Multiple integration and the use of a flawed selection marker can be used together to produce stable, high-level expression of a heterologous protein (Thak et al., 2020).

Multiple steps are required for the secretion of recombinant proteins in yeast, including translocation to the ER and protein folding. The selection of a secretion SP sequence is essential for achieving high levels of protein secretion. However, the efficacy of different SP sequences can vary significantly based on the protein's sequence, structure, and expression regulation. A hierarchical assembly method was utilized to generate diverse secretion libraries, and in silico analysis based on the D-score of a given sequence segment was utilized to efficiently select endogenous SPs. Engineering the protein secretion pathway by doing things such as overexpressing chaperones or redox enzymes can also make it easier for recombinant proteins to fold and get out of the cell (Thak et al., 2020). Recent research has implemented the push-and-pull concept, which entails the overexpression of the most relevant factors of the Hsp70 chaperone cycles in the cytosol to enhance translocation competency. Hsp70 chaperones engage in catalytic cycles with their substrates, associating and dissociating in a regulated manner. Co-chaperones Hsp40 facilitate the transition to the state of high affinity (Zahrl et al., 2022).

In addition, toolboxes for promoter and transcriptional terminator engineering have been developed to improve protein expression in *P. pastoris*. Natural, strong, and tightly regulated promoters, as well as short artificial promoters for fine-tuning gene expression, have been characterized as inducible and constitutive promoters of varying strengths. Synthetic core promoters have also been developed to aid in the development and application of novel orthogonal promoters. Other inducible or constitutive promoters are preferred over the toxic methanol inducible system for optimizing *P. pastoris* cell factories. In addition to promoters, transcriptional terminators play an important role in gene expression regulation. To regulate mRNA stability and gene expression levels in yeast, synthetic terminators have been successfully designed and characterized. A genome engineering toolbox has also been developed to enable efficient and repeated genome operation and the construction of marker-free strains. To improve gene targeting efficiency, the Cre-loxP system from *S. cerevisiae* was successfully introduced into *P. pastoris* (Kang et al., 2017). Furthermore, genetic switches can be used to improve transcriptional or translational expression performance. In yeast, endogenous systems, such as the Gal4 transcriptional switch, have been extensively utilized. Additionally, researchers can use heterologous, ligand-responsive DNA-binding proteins to create synthetic genetic switches. Due to their inappropriate switching properties, however, the majority of genetic switches cannot be used directly in synthetic biology applications. For instance, their sensitivity or specificity to the desired chemical may be low (Tominaga et al., 2022).

According to their mode of regulation, yeast synthetic transcription switches can be divided into two groups: activator mode and repressor mode. In the activation mode, yeast can be used to generate synthetic transcriptional switches by fusing ligand-responsive DNA-binding proteins, such as bacterial transcription factors (bTFs), with synthetic transcriptional activators (sTAs). Using suitable design parameters, the ligand-dependent binding of bTFs to their operator DNA sequences can be translated into the expression of genes. The Tet-OFF system, which employs a synthetic transcription activator and has been extensively utilized in synthetic biology projects, is a well-known and proven technology. Using different TetR homolog transcription repressors and their cognate operator DNAs and ligands, researchers have created a series of "inducer-OFF" genetic switches that function in *S. cerevisiae*. On the basis of bacterial transcriptional corepressors and activators and ligand-dependent nuclear localization, more examples of sTAs have been made (Tominaga et al., 2022). Additionally, some sTAs can be generated without a eukaryotic activation motif by employing bacterial LysR-type transcriptional regulators (LTTRs) that can activate transcription in *S. cerevisiae* even without the fusion of a eukaryotic activation motif. In the repressor mode, yeast can produce synthetic transcriptional switches by fusing ligand-responsive DNA-binding proteins with engineered transcriptional repressors (eTRs). Similar to activation mode switches, ligand-dependent DNA-binding protein binding to their operator DNA sequences can inhibit gene expression. General and specific strategies to improve the performance of each type of yeast transcriptional switch include optimizing the design of sTAs and eTRs to increase sensitivity and specificity to target chemicals, as well as enhancing gene expression control through feedback loops (Tominaga et al., 2022).

Recently, a Golden Gate-based modular cloning system called GoldenPiCS was developed for the genetic engineering of *P. pastoris*. It consists of three levels of hierarchical backbone that permit the flexible generation of overexpression plasmids. Individual components, such as promoters, coding sequences, and transcription terminators, are assembled using Golden Gate Assembly at the lowest level of cloning. Golden Gate plasmids contain multiple transcription units simultaneously, allowing screening and transformation to occur in only 9 days. The selection of promoters and terminators was based on the transcriptional regulation and expression level of their natively regulated genes. Under the appropriate conditions, the strength and regulation of the selected promoters and terminators were characterized using the intracellular reporter eGFP. *P. pastoris* was used to episomally maintain the assembled plasmids. GoldenPiCS's availability on Addgene has made it a widely accessible tool for *P. pastoris* strain engineering (Prielhofer et al., 2017).

A synthetic yeast platform based on *P. pastoris* (SynPic-X) that can precisely respond to user-defined signals was also recently developed. The platform is intended to increase the production capacity of recombinant proteins and peptides by boosting *P. pastoris* transcription levels. To do this, a synthetic transcription device called iTSAD was developed. Exogenous DNA binding proteins (DBPs) and binding sequences (BSs), as well as the activation domains (ADs) of endogenous transcription factors, activate this device (TFADs) (Q. Liu, Song, et al., 2022). The ADs of the *P. pastoris* transcription activators Mit1, Mxr1, and Prm1 were fused to the C terminus of the *E. coli* LacI protein to generate three chimeric activators. With respect to the l*acO*-cP AOX1 input promoter, the LacI-Mit1AD chimeric activator exhibited the strongest activation effect. The researchers incorporated CRISPRi-mediated transcriptional repression of iTSAD to gain greater control over the iTSAD system. To achieve the desired repression, a CRISPRiD encoding a human codon-optimized dCas9 protein was designed. Using iTSAD, the researchers determined the relationship between the input strength of giRNA F1 and the output strength after amplification. They discovered that a weak expression of giRNA F1 could significantly inhibit the output of iTSAD, and that a strong input promoter, PGAP, 50-fold inhibited the output (Liu, Song, et al., 2022). CRISPRiD was subjected to RNA-interaction-mediated derepression regulation to further enhance iTSAD control. Two trigger RNAs were designed to interact with the giRNA and six activator RNAs to disable CRISPRiD by inhibiting the DNA-targeting or Cas handle regions of the giRNA. Two orthogonal CRISPR regulatory proteins, VRER and dCpf1, were chosen to avoid cross talk between CRISPRa and CRISPRi. Thus, three gaRNAs and three craRNAs were designed for activation, and P GAP-driven activator RNAs were used to achieve CRISPRiD repression interference. Even at high activator RNA expression levels, the derepressive effect was found to have a relatively limited effect (Liu, Song, et al., 2022). Finally, it was attempted to antagonize CRISPRiD using CRISPRaD. They created chimeric activators by fusing dCas9 with Mit1AD, Mxr1AD, and a viral activator that is codon-optimized. They designed gaRNAs and craRNAs to activate CRISPRaD and discovered that the gaRNA/craRNA-based CRISPRaD system activated iTSAD effectively (Liu, Song, et al., 2022).

Yeast Episomal Vectors (YEp) are an additional expression system that has been utilized for the expression of recombinant proteins in yeast and could potentially be applied to AMPs. These vectors contain multiple cloning sites, including unique restriction enzyme sites. Additionally, they contain multiple antibiotic-resistance genes, allowing scientists to select yeast transformants with ease. The YEp vectors offer a number of benefits over conventional yeast expression vectors. They are simpler to prepare than pYES2 and pYES2-based vectors, and they facilitate more uniform protein expression. Furthermore, they are useful for transient and stable protein expression (Gnügge & Rudolf, 2017). The YEp vectors offer the following advantages over conventional yeast expression vectors:

- Provide multiple cloning sites, including unique restriction enzyme sites.
- Contain multiple antibiotic-resistance genes, allowing researchers to easily select yeast transformants.
- Are easier to prepare than conventional yeast expression vectors.
- Allow more uniform protein expression than pYES2 and pYES2-based vectors.
- Are useful for both transient and stable expression of proteins and peptides.

The preparation of YEp vectors involves first digesting the vector with restriction enzymes and then ligating the appropriate insert into the vector. It is recommended that vectors be cut with *Bgl*II and *Xba*I for expression vectors. For dual expression systems, *Spe*I and *Bst*XI are recommended for cutting vectors. Refer to the manuals for commercial Golden Gate cloning kits and Quick clone kits for documentation on how to perform the digestion and ligation reactions. YEp vectors are compatible with cDNA, PCR products, PCR fragments, cosmid inserts, genomic clones, and phage clones, among other types of DNA inserts. It is recommended that when expressing a protein or peptide from YEp vectors, a relatively potent promoter be used to ensure that the desired protein is expressed at sufficient levels. It is also recommended to use a full-length version of the gene encoding the protein to prevent the loss of essential regulatory sequences present in the truncated version (Gnügge & Rudolf, 2017).

Due to the low copy number of episomal vectors in *Y. lipolytica*, chromosomal autonomously replicating sequence/centromere replication origins are utilized. Using a hybrid promoter, Yeastern Biotech offers the commercial expression vectors pYLEX1 and pYLSC1 https://www.cosmobiousa.com/content/document/yeastern/Yeastern-catalog2017.pdf. The *S. cerevisiae* SUC2 gene and auxotrophic markers URA3 and LEU are selection markers. Endogenous SPs from *Y. lipolytica* XPR2 and LIP2 genes are frequently used for secretion. For molecular studies, *S. pombe* possesses a variety of plasmids, including episomes, expression vectors, epitope-tagging plasmids, and integration vectors. The *ura4* gene and the leu1 gene are frequently used selection markers, whereas the strong Padh1 and the PCaMV variant are frequently used constitutive promoters. In relation to its two peptide pheromones, fission yeast was found to have two secretion pathways. *Candida utilis* is important for SCP production and uses integrative vectors based on multiple chromosomal loci frequently. As selection markers, L41 and auxotrophic markers are utilized, and P TDH3, P PGK1, and P PDC1 are common promoters. Since it lacks natural plasmids, *Kluyveromyces marxianus* requires new expression vectors derived from other yeast species (Gündüz Ergün et al., 2019).

Due to *Y. lipolytica*'s enormous potential for industrial applications, metabolic engineering, and synthetic biology strategies have been implemented to develop new strains to expand its industrial use. Utilizing CRISPR/Cas9 tools and new modeling techniques has been essential for these advancements. The ability of *Y. lipolytica* to transit through the acetyl-CoA and malonyl-CoA pathways provides a significant advantage for the production of diverse heterologous products, as does the organism's capacity to utilize diverse carbon substrates. The transformation of *Y. lipolytica* from a nonconventional yeast to a common yeast necessitates an enhanced set of genetic and synthetic tools. New synthetic control elements have been developed to enable straightforward engineering, despite the fact that initial transformation strategies remain limited. It has been demonstrated that hybridization strategies are effective for generating robust, constitutive gene expression. The application of CRISPR/Cas9 systems has emerged as a straightforward method for enhancing engineering efforts, particularly in targeted markerless gene integration and CRISPR-mediated gene repression. Standardization efforts have produced a set of toolboxes for cloning vectors and integrating genes into *Y. lipolytica*, allowing for rapid pathway testing. The development of new synthetic promoters and terminators will further enhance *Y. lipolytica*'s ability to regulate gene expression. In addition, inducible promoters can assist in decoupling the growth and production phases in industrial fermentation, whereas metabolite-responsive elements can expand the inducibility spectrum to include alkanes and erythritol (Markham & Alper, 2018).

Despite the fact that *S. cerevisiae* and *P. pastoris* are the most established conventional model yeasts, the use of nonconventional yeast strains as expression systems has become an attractive alternative to conventional model yeasts due to their unique characteristics and potential advantages. In particular, *K. marxianus* and *K. lactis* have gained interest as potential platforms for bioproduction due to their ability to produce diverse molecules, such as lipids and organic acids. To optimize protein production in nonconventional yeast strains, it is necessary to consider factors such as promoter selection, codon optimization, SP design, and media composition. In addition, the application of various genetic engineering techniques, such as gene knockouts, CRISPR-Cas9-mediated genome editing, and heterologous expression of enzymes, can further improve protein expression and secretion. As more and more expression systems for nontraditional yeasts are made, it is becoming clear that these strains have the potential to completely change the field of bioproduction because they are better than traditional model organisms (Patra et al., 2021).

3 Cell-free synthesis: Revolutionizing biotherapeutic manufacturing

E. coli remains the most economical method for producing numerous complex proteins, such as insulin and antibodies. For proteins requiring complex assembly and posttranslational modifications, other synthesis systems, such as those in yeast and mammalian cells (such as glycosylation and lipidation), are used (Cavicchioli et al., 2019). In vitro translation systems were developed as a result of the early demonstration that cell integrity is not required for protein synthesis. A crude lysate from any organism (which provides the translational machinery, accessory enzymes, tRNA, and factors) is combined with an exogenously added RNA template, amino acids, and energy source to accomplish translation in its most fundamental form. In contrast to the "coupled" or "combined" transcription and translation configuration in which the mRNA is transcribed in situ from an added DNA template, this classical in vitro translation scheme is referred to as "uncoupled" (Jermutus et al., 1998). Coupled systems have higher protein yields and are simpler and quicker to operate than uncoupled systems. They do, however, require the addition of additional NTPs and a highly processive RNA polymerase, such as those encoded by T7, T3, or SP6 bacteriophages. Numerous new applications have emerged as a result of the use of plasmid or PCR templates as opposed to purified mRNAs (Fig. 7.1).

Cell-free synthesis (CFS), also known as cell-free transcription and translation, is the addition of nucleotides, amino acids, metabolic intermediates, and salts to cellular components (either a cell lysate or purified recombinant elements) to produce a nucleic acid or protein from a genetic template added to the reaction. In the past decade, academic and commercial interest in this innovative technology has grown substantially (Dopp et al., 2019). This motivation is primarily derived from the potential to democratize access to the machinery of biology by eliminating the need to genetically modify cells (Melinek et al., 2020). CFS has the potential to revolutionize healthcare in a manner analogous to how personal computers transformed the information technology industry. CFS platforms are increasingly recognized as a potential enabling technology for stratified approaches to facilitate the decentralized manufacturing of biological products (Adiga et al., 2018). Although it is still in its developmental stages as a manufacturing technology, CFS offers greater flexibility and possibly greater robustness than cell-based biotherapeutic manufacturing. Some examples of CFS use include the production of mutant proteins (Spirin, 2004), the development of cell-free biosensors (Zhang et al., 2020), and the control of posttranscriptional processes such as alternative splicing (Mayeda et al., 1992) or cotranslational folding (Moreno Ayala et al., 2020).

While CFS has many potential biotechnology applications, it is also a highly promising platform for drug discovery and development. Using CFS, scientists can create biologically active molecules on a large scale with greater precision and at a lower cost than conventional drug discovery techniques. CFS's achievements are primarily attributable to advances in reengineering the system to produce more complex proteins or highly hydrophobic peptides and to increase production yields (2 mg/mL in batch format (Caschera & Noireaux, 2014) and 6 mg/mL in fed-batch format (Salehi et al., 2016)) while lowering reagent costs for lysate-based systems (as low as $1-$2/mL for the least expensive batch system based on *E. coli* lysate when prepared in-house (Bundy et al., 2018)). Because only biomanufacturing can reliably synthesize large proteins and highly hydrophobic molecules, and therapeutic proteins and peptides have a particularly high value ($100/mg–$1000/mg) (Ecker et al., 2015), cell-free biomanufacturing of therapeutic proteins has received the most attention. Some examples of AMPs produced using CFS technology include Magainin-2, a naturally occurring AMP found in the skin of frogs that has potent activity against Gram-negative bacteria and fungi (Lee et al., 2010), LL-37, which is a human cathelicidin AMP that has potent activity against a wide range of pathogens, including bacteria, fungi, and viruses (Wong et al., 2018), Plectasin, a naturally occurring AMP found in fungi against a variety of pathogens including methicillin-resistant *Staphylococcus aureus* (MRSA) (Hara et al., 2008), and Indolicidin, a naturally occurring AMP found in neutrophils that has potent activity against Gram-negative bacteria and fungi (Albiol Matanic & Castilla, 2004).

The cell-free gene expression reaction has three parts: the cell extract, a reaction mixture, and a mixture of DNA and inducers that contains the genetic instructions for the reaction (see Fig. 7.1).

- *Extract*: Ribosomes, RNA polymerase, and other transcription and translation accessory proteins (such as sigma factors, initiation factors, and elongation factors) derived from a source strain are included in the extract. There are also metabolic enzymes for energy and cofactor regeneration. Frequently, the source strain's genome has been optimized to reduce protease and nuclease activity. To obtain a functional extract, the source strain is commonly grown in enriched media and lysed by homogenization, sonication, bead-beating, or freeze-thawing. Depending on the application, additional postlysis purification steps, such as ribosomal runoff reaction and dialysis, may be performed (Michel-Reydellet et al., 2004).
- *Reaction mix*: This mixture of supplemented cofactors for protein synthesis includes amino acids, nucleotides, salts (such as ammonium, magnesium, and potassium cations), polyamines (such as putrescine, spermidine), an energy source (such as sugars or polysaccharides, glycolytic intermediates, or creatine phosphate), molecular crowding agents (such as polyethylene glycol or Ficoll), metabolic cofactors (such as nicotin (for example, oxalate to inhibit gluconeogenesis, glutathione for disulfide bond formation, and folinic acid for formylmethionine synthesis) (Jewett et al., 2008; Jewett & Swartz, 2004a, 2004b).
- *DNA template*: In the reaction, the template used in the reaction can be plasmid DNA or a linear expression template obtained through PCR, typically at a concentration of 1–20 nM.

The three components are incubated together at temperatures ranging from 16°C to 37°C, and batch protein synthesis reactions typically take 1–24 h (Sun et al., 2013). The standard reaction volume is 10 L, but the reaction can be carried out in 1 mL tubes or on the bottom of well plates to increase the oxygenated headspace and thus facilitate oxidative phosphorylation (Shrestha et al., 2012; Sun et al., 2013). Depending on the application, the reaction volume can vary by orders of magnitude. Protein yields in a reaction are typically in the hundreds of micrograms per mL range, with protein production in conventional batch reactions exceeding 2 mg/mL for model proteins (Caschera & Noireaux, 2014). Higher yields can be obtained by extending the reactions through the continuous exchange of reaction by-products and a nutrient feed stream in an ultrafiltration cell, dialysis cassette, or microfluidic pumping (Khnouf et al., 2009). These yields are sufficiently high for conventional reporter proteins, such as green fluorescent protein (GFP) or its variants, that gene expression can be measured across several orders of magnitude.

3.1 Biosafety

A component of biosafety called biocontainment deals with the physical confinement of organisms and species that might be harmful to ecology and human health to prevent their release into the general population. Many present biocontainment techniques may not be revised for today's concerns, despite their widespread use and significant triumphs, particularly when it comes to the release and spread of new transgenes and/or transgenic organisms (Lee et al., 2018). Therefore, it is imperative to create new technologies to address the biocontainment hazards that jeopardize biosafety and biosecurity. Several fresh approaches that employ cell-free systems have been suggested among the numerous others that have been put out so far (Lee et al., 2018). The advantage of obtaining proteins using cell-free expression systems is that they stay abiotic, without the majority of the biotic processes that generally occur in cells while containing genes and DNA that would be proliferating, replicating, and changing. Using these systems reduced the biosafety risk in all circumstances compared with using live systems, a frequently overlooked characteristic.

3.2 High-throughput and on-demand production

Following the pioneering reports on the direct use of DNA polymerase chain reaction (PCR) products for programming protein synthesis in cell-free transcription-translation systems (Lesley et al., 1991; Martemyanov et al., 1997; Ohuchi et al., 1998), functional mapping of genomes through the expression of genomic libraries and the construction and screening of protein libraries became one of the cell-free technology's most essential applications (Busso et al., 2003; Endo & Sawasaki, 2004; Rungpragayphan et al., 2002; Sawasaki et al., 2002). On the one hand, we have the cell-free translation of large encoding libraries that allows researchers to explore diverse phenotypic peptide sequence spaces. On the other hand, the high-throughput screening approaches allow the examination of interactions between numerous chemical or biological compounds and particular targets in a reliable, speedy, and highly reproducible manner. For these tasks, a multiwell format for cell-free systems and robotic process automation have been proposed (Merk et al., 2003; Miyazaki-Imamura et al., 2003).

The combination of increasingly rapid gene synthesis technologies and shelf-stable on-demand protein synthesis is a potent instrument for high-throughput protein production. However, the current understanding of gene product structure and function needs to catch up to gene sequence data (Murray & Baliga, 2013; Takahashi et al., 2015). Researchers attempting to comprehend proteomes and peptidomes are notably aided by methods for high-throughput and cost-effective peptide screening. Additionally, high-throughput peptide synthesis is required for rational peptide engineering and directed protein evolution.

Due to the fact that typical cell-free extracts must be used immediately or frozen, and preparation of the extracts takes several hours, reliance on these extracts can potentially delay the execution of rapid peptide expression cycles. On-demand peptide expression systems, which transcribe and translate from rapidly produced DNA and provide activity assessment without product purification, permit the execution of parallel synthesis cycles in as little as 3–4 h and facilitate activity assessment without product purification (Murray & Baliga, 2013). Peptides can also be produced on demand using lyophilized cell-free reactions, increasing the flexibility and efficiency of chemical supply pipelines. In the chemical industry, biocatalysts are at the forefront of "green chemistry," eliminating harsh chemical catalysts and waste streams while reducing thermal energy costs. Thanks to quickly synthesizable biocatalysts, chemical supply pipelines can be modified in response to shifting demand for a chemical commodity. Thus, the chemical feedstock may be utilized more efficiently to produce the most profitable chemical product.

4 Exploring the drivers and challenges of biological antimicrobial production

4.1 Motives for utilizing a biological approach for the production of AMPs

The use of biological production approaches, including recombinant production, has become increasingly popular in the production of AMPs. In this section, we will talk about what makes it important to use biological production methods to make AMPs.

- Driver 1: Cost-effectiveness: Recombinant production methods are more cost-effective than traditional chemical synthesis methods due to their ability to produce large quantities of AMPs in a short period of time (Gaglione et al., 2019).
- Driver 2: Sustainability: Biological production methods, such as recombinant production, are a better way to make antibiotics because they use renewable resources and make less waste, which lowers the risk of contamination and the impact on the environment (Freudenberg et al., 2021).
- Driver 3: Consistency: The use of recombinant production methods allows for the standardization of the production process, which reduces the risk of variation in quality. This consistency in quality is important to make sure that the AMPs are safe and effective for both people and animals (Zhang et al., 2003).
- Driver 4: Scalability: Biological production methods are scalable, allowing companies to produce AMPs in large quantities, making them more accessible and affordable to consumers. This can help meet the growing need for new antibiotics and help stop bacteria from becoming resistant to antibiotics (Rademacher et al., 2019; Ramírez et al., 2020).

4.2 Challenges

4.2.1 Proper folding and posttranslational modifications

The bioactivity of recombinant AMPs is dependent on proper folding and posttranslational modifications. AMPs are frequently synthesized as linear polypeptides, and the bioactivity of these molecules is determined by their three-dimensional structure. Because of their nonspecific targeting of both bacterial and eukaryotic cell membranes, improper folding can result in loss of antimicrobial activity or even toxicity (Baneyx & Mujacic, 2004). Posttranslational modifications (PTMs), on the other hand, can play an important role in determining the bioactivity of recombinant AMPs. PTMs include, for example, phosphorylation, glycosylation, and disulfide bond formation. These modifications may be critical for the peptide's stability, solubility, and activity (Jenkins et al., 2008). As a result, it is critical to ensure that recombinant production methods allow for the appropriate PTMs to occur to maintain the peptide's bioactivity. Several approaches have been developed to achieve proper folding and PTMs of recombinant AMPs. Coexpression with chaperone proteins, for example, or fusion with carrier proteins, can improve peptide solubility and promote proper folding (Reyes et al., 2017; Thomas et al., 1997). Alternatively, specific PTMs can be introduced into the peptide through chemical synthesis or enzymatic modification (Gomord & Faye, 2004).

4.2.2 Toxicity to the host organism used for production

The potential toxicity to the host organism used for production is one of the major concerns in the recombinant production of AMPs. AMPs are known to have broad-spectrum activity against a variety of microorganisms, and this activity can be harmful to the host organism that produces the peptide. This toxicity can result in impaired cell growth, decreased expression levels, and cell death. Several strategies have been developed to avoid toxicity. One of the most common approaches is to use expression systems that are less sensitive to the peptide's cytotoxic effects. The use of bacterial systems such as *E. coli*, for example, can be problematic due to their susceptibility to the peptide's antimicrobial activity. On the other hand, yeast or filamentous fungi are less sensitive to the toxicity of AMPs and can provide a better platform for the production of these peptides. Another approach is to use inducible expression systems, which allow for precise control over peptide expression. This can help reduce toxicity by limiting the host organism's exposure to the AMP. Similarly, the strategies described at the beginning of this chapter are ideal for dealing with this problem.

4.2.3 Stability and shelf-life of the final product

AMPs are frequently sensitive to environmental factors such as temperature, pH, and humidity, which can have a significant impact on their stability and activity (McClements, 2018). As a result, developing stable and robust formulations that can withstand these conditions during storage and transportation is critical. Various strategies can be used during the manufacturing process to achieve stability, including the use of stabilizing agents, the optimization of formulation conditions, encapsulation, immobilization on nanoparticles, and lyophilization (freeze-drying) of the final product. By providing a protective environment, stabilizing agents such as sugars, amino acids, and surfactants can help shield the peptide from degradation (Fujimoto et al., 2008; Tacias-Pascacio et al., 2020). Optimization of formulation conditions such as pH, ionic strength, and temperature can also contribute to the storage stability of peptides (Frokjaer & Otzen, 2005; Wang, Kumru, et al., 2013). Peptides can also be made more stable by encapsulating them in micro- or nano-sized carriers such as liposomes, polymeric nanoparticles, and biopolymers. The most common encapsulation materials consist of natural polymers such as alginate, chitosan, and cellulose derivatives, synthetic polymers such as PLGA and PEG, and lipids such as cholesterol (Alvarado et al., 2019; Garcia-Brand et al., 2022; Han et al., 2022). Immobilization on nanoparticles provides additional benefits, including enhanced solubility and drug loading, increased bioavailability, prolonged duration of action, and reduced immunogenicity (Torres-Vanegas et al., 2022, 2021). Recombinant AMPs are frequently stabilized, and their shelf life is lengthened by lyophilization. It involves removing the sample's water, thereby preventing deterioration caused by humidity. In addition, the freeze-dried product is easy to transport and store, reducing the likelihood of peptide damage during transport.

4.2.4 Regulatory hurdles for approval and commercialization

The commercialization of any therapeutic product, including AMPs produced through recombinant technology, requires overcoming regulatory obstacles. Regulatory agencies such as the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA) have stringent requirements for the safety, efficacy, and quality of therapeutic products, and obtaining regulatory approval can be a time-consuming and expensive endeavor. AMPs are required to demonstrate their safety profile as a primary regulatory obstacle. Since AMPs are typically derived from naturally occurring proteins, they are regarded as generally safe. However, the use of recombinant technology in their production may result in unanticipated modifications that compromise their safety. Therefore, extensive safety testing must be conducted to ensure that the manufactured AMPs are safe for human or animal consumption. AMPs must also demonstrate their efficacy to pass regulatory muster. Due to the fact that AMPs can have a wide variety of targets and mechanisms of action, their efficacy must be evaluated for each specific indication. To demonstrate efficacy, safety, and appropriate dosing regimens, rigorous preclinical and clinical testing is necessary. In addition, regulatory agencies may require comparability studies to demonstrate that the recombinantly produced AMP is structurally, functionally, and biologically equivalent to the naturally occurring peptide.

5 Future perspectives

Despite their potential, producing AMPs on a large scale remains a daunting challenge. Traditional methods for producing AMPs—chemical synthesis and recombinant expression in bacteria, yeast, fungi, insect cells, mammalian cells, and plants—have several limitations, such as low yields, high costs, and complex purification processes. As such, there has been an increasing interest in exploring novel biologically based expression systems for AMPs that can circumvent these issues. Other than novel expression systems, there are other important factors that must be considered to ensure the development and successful application of these antimicrobial agents. These include optimizing AMP production, protein folding, and posttranslational modifications, as well as safety evaluation and in vivo activity. They also help overcome regulatory and commercialization hurdles. These multidimensional issues are essential to advancing AMP research and development. They also contribute to the global effort to fight antimicrobial resistance and open the door to innovative and effective treatments for infectious diseases.

5.1 Novel expression systems for AMPs

In this section, we discuss three potentially novel biologically based expression systems for AMPs.

5.1.1 Microalgae

Microalgae are photosynthetic microorganisms that have recently gained attention as potential expression systems for recombinant proteins (Malla et al., 2021). Microalgae possess several advantages as a production platform, including rapid growth rates, the ability to utilize light energy and carbon dioxide, and the potential for large-scale cultivation in outdoor systems. Additionally, microalgae are generally recognized as safe (GRAS) and are not known to harbor human pathogens, reducing the risk of contamination in the final product (Wong et al., 2022). However, microalgae expression systems also present several challenges. First, the development of efficient transformation and selection methods is still in its infancy (Grama et al., 2022). Second, the downstream processing of AMPs from microalgae may be complex due to the presence of various cell wall components and other intracellular components (Rahman et al., 2022). Nonetheless, with continued research and optimization, microalgae could become a promising platform for the production of AMPs.

5.1.2 Protozoa

Protozoa are unicellular eukaryotic organisms that have not been extensively explored as expression systems for recombinant proteins. However, they have several characteristics that make them a potential candidate for AMP production. Protozoa can grow rapidly, have simple nutritional requirements, and can be cultivated in suspension, facilitating large-scale production (Niimi, 2011). One of the primary challenges of using protozoa as an expression system is the lack of established genetic engineering tools and protocols. Moreover, some protozoa are known to be human pathogens, raising concerns about the safety of using them as expression hosts. Further research is required to identify suitable protozoan species for AMP production, develop genetic engineering techniques, and address safety concerns.

5.1.3 Bacteriophages

Bacteriophages, commonly referred to as “phages,” are viruses that infect bacteria. Phages have been extensively researched for their potential use in phage therapy—a method to fight bacterial infections (Lin et al., 2017). Recently, there has been increased interest in exploring the potential of phages as expression systems for AMPs. Phages can be engineered to display AMPs on their surface or produce them within bacteria upon infection (Davydova, 2022). Phage expression systems for AMPs offer several advantages, such as high specificity toward target bacteria, self-replication, and rapid evolution. However, this approach also comes with its share of challenges. For instance, the expression of AMPs within bacteria may lead to cell lysis, decreasing production yields (Egido et al., 2023). Furthermore, the regulatory framework for using bacteriophages in therapeutic applications is still being developed and could potentially restrict the adoption of this technology (Fauconnier, 2019).

5.1.4 Trade-offs and challenges

In developing novel biologically based expression systems for AMPs, it is crucial to consider the trade-offs involved in balancing different factors. For instance, while some expression systems may offer high production yields, they may also be associated with complex downstream processing or safety concerns. Similarly, the use of photosynthetic organisms such as microalgae may reduce production costs by utilizing sunlight and CO_2, but the development of efficient genetic engineering tools remains a challenge. Furthermore, each expression system comes with its own set of challenges. For microalgae, optimization of transformation and selection methods is required, while protozoa demand the development of genetic engineering tools and protocols as well as addressing safety concerns. In the case of bacteriophages, balancing the production yield with potential host cell lysis and navigating regulatory frameworks are key challenges.

5.1.5 Environmental and ethical considerations

When making decisions regarding novel biologically based expression systems for AMPs, it is essential to take into account both the environmental and ethical impacts of each approach. For instance, using microalgae as an expression system could contribute to carbon sequestration and reduce greenhouse gas emissions; however, large-scale cultivation in open systems could contaminate natural water bodies, potentially impacting local ecosystems (Wimmerova et al., 2023).

Similar to protozoa or bacteriophages as expression systems, using them for expression may raise ethical issues if they involve the release of genetically modified organisms into the environment. Therefore, it is essential to develop containment strategies and conduct thorough risk assessments to minimize potential negative impacts on ecosystems and human health.

5.2 In vivo activity and safety

It is crucial to evaluate the in vivo activity of AMPs to develop effective antimicrobial treatments. In vitro tests, such as the determination of the minimum inhibitory dose (MIC) or time-kill assays, can provide preliminary insights into the antimicrobial properties of AMPs. These assays cannot predict in vivo efficacy because they don’t account for factors such as pharmacokinetics or protein stability. For in vivo testing of AMPs, scientists can use a variety of animal models, such as murine, zebrafish, or insect models. These models allow for the evaluation of AMPs’ efficacy in clearing the target organism, their impact on host life, and potential synergistic or antagonistic interactions. Additional in vivo imaging methods, such as fluorescence imaging or bioluminescence imaging, provide real-time insights into infection dynamics and the distribution of AMPs in the host.

It is essential to ensure the safety of AMPs used for therapeutic purposes. This involves evaluating their potential to be immunogenic, toxic, and induce off-target effects in relevant animal models, and then, ultimately, in human clinical trials. It is possible to assess the potential immunogenicity of AMPs by monitoring the host’s immune response to the peptides. This includes the production of antipeptide antibodies and the activation of immune cells. Toxicity assessment can include assessing the effect of AMPs upon host cell viability, organ function, overall health, and determining the Therapeutic Index, which is the ratio between the toxic dose and the effective dose. Research can help ensure that effective, safe, and clinically proven antimicrobial therapies are developed by thoroughly examining the in vivo activity of AMPs.

6 Conclusion

The rising prevalence of antibiotic resistance has highlighted the urgent need to find novel antimicrobial strategies. These include the development and optimization of biologically based expression systems for AMPs. We have examined the potential of microalgae/protozoa and bacteriophages for the production of AMPs. They are a promising alternative to traditional expression systems such as bacteria, yeast, fungi, or insect cells. Each of these innovative expression systems provides unique advantages and potential for the production of AMPs. Because of their rapid growth, ability to harness carbon dioxide and light energy, and compatibility for large-scale outdoor cultivation, microalgae could be a cost-effective and long-lasting option. While relatively new, protozoa have the potential to grow quickly and require very few nutritional requirements. They can also be easily grown in suspension culture.

Bacteriophages provide high specificity, self-repetition, and the potential for rapid evolution, which makes them attractive candidates for AMP manufacturing. The downside is that embracing novel expression systems can present its own challenges and trade-offs. To make microalgae more productive, genetic engineering tools are needed. The identification of appropriate protozoan species and safety concerns are critical in the case of protozoa. This is why genetic engineering techniques have been developed. Issues facing bacteriophages are production yields and managing the regulatory landscape of phage-based therapeutic applications. It is important to think about the ethical and environmental impacts of these new expression systems. Microalgae could, for instance, help reduce greenhouse gas emissions by storing carbon, while large-scale cultivation could negatively impact local ecosystems.

In addition, genetically modified prozoa and bacteriophages can pose ethical issues and require containment strategies to reduce their potential adverse effects on ecosystems and human health. Continued research and interdisciplinarity are crucial to overcoming these difficulties and unlocking the full potential of these innovative expression systems. If we invest in the development of protozoa and microalgae-based platforms, it will be possible to speed up the production of new antimicrobial strategies to counter the increasing threat of antibiotic resistance. By doing so, we can improve global health outcomes and promote a more sustainable, responsible approach to biopharmaceutical product production. Researchers must also explore new expression systems. They must address key aspects such a optimizing AMP production, protein folding, and posttranslational modification, as well as ensuring safety and in vivo activity. Optimizing AMP manufacturing requires the selection and design of appropriate expression hosts.

Researchers can enhance the therapeutic power of these promising antimicrobial agents by maximizing their yield and quality. Protein folding and posttranslational modifications are crucial in determining the bioactivity as well as the stability and functionality of AMPs. For AMP-based therapies that are effective, it is vital to ensure proper folding and modifications of the expression host. Researchers have many options to tackle these issues, including coexpression with chaperone proteins, genetic engineering in host organisms, and the use of cell-free synthesis systems. For the development of safe and effective antimicrobial medicines, it is essential to evaluate the safety of AMPs and their in vivo activity. To ensure that in vivo safety is maintained, researchers must use suitable animal models for infection. These important factors, along with the development of novel expression platforms, can all be used by researchers to contribute to the success of antimicrobial therapies. This holistic approach to AMP development is crucial for the ongoing battle against antibiotic resistance and the pursuit of innovative solutions to the global healthcare crisis.

7 AI disclosure

During the preparation of this work, the author(s) used GPT 4 as style and grammar checker throughout the chapter. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

Adiga, R., Al-Adhami, M., Andar, A., Borhani, S., Brown, S., Burgenson, D., Cooper, M. A., Deldari, S., Frey, D. D., Ge, X., Guo, H., Gurramkonda, C., Jensen, P., Kostov, Y., LaCourse, W., Liu, Y., Moreira, A., Mupparapu, K. S., Peñalber-Johnstone, C., … Zuber, A. (2018). Point-of-care production of therapeutic proteins of good-manufacturing-practice quality. *Nature Biomedical Engineering*, *2*(9), 675–686. https://doi.org/10.1038/s41551-018-0259-1.

Ahmad, M., Hirz, M., Pichler, H., & Schwab, H. (2014). Protein expression in Pichia pastoris: Recent achievements and perspectives for heterologous protein production. *Applied Microbiology and Biotechnology*, *98*(12), 5301–5317. 0175-7598 https://doi.org/10.1007/s00253-014-5732-5.

Albiol Matanic, V. C., & Castilla, V. (2004). Antiviral activity of antimicrobial cationic peptides against Junin virus and herpes simplex virus. *International Journal of Antimicrobial Agents*, *23*(4), 382–389. https://doi.org/10.1016/j.ijantimicag.2003.07.022.

Alvarado, Y., Muro, C., Illescas, J., Díaz, M.d. C., & Riera, F. (2019). Encapsulation of antihypertensive peptides from whey proteins and their releasing in gastrointestinal conditions. *Biomolecules*, *9*(5). 2218273X. MDPI AG, Mexico. Available from: https://www.mdpi.com/2218-273X/9/5/164/pdf. https://doi.org/10.3390/biom9050164.

Amparyup, P., Kondo, H., Hirono, I., Aoki, T., & Tassanakajon, A. (2008). Molecular cloning, genomic organization and recombinant expression of a crustin-like antimicrobial peptide from black tiger shrimp Penaeus monodon. *Molecular Immunology*, *45*(4), 1085–1093. 01615890 https://doi.org/10.1016/j.molimm.2007.07.031.

Andersson, L., Blomberg, L., Flegel, M., Lepsa, L., Nilsson, B., & Verlander, M. (2000). Large-scale synthesis of peptides. *Biopolymers*, *55*(3), 227–250. 0006-3525 https://onlinelibrary.wiley.com/doi/abs/10.1002/1097-0282%282000%2955%3A3%3C227%3A%3AAID-BIP50%3E3.0.CO%3B2-7.

Anné, J., Maldonado, B., Van Impe, J., Van Mellaert, L., & Bernaerts, K. (2012). Recombinant protein production and streptomycetes. *Journal of Biotechnology*, *158*(4), 159–167. 01681656 https://doi.org/10.1016/j.jbiotec.2011.06.028.

Arnau, J., Yaver, D., & Hjort, C. M. (2020). Strategies and challenges for the development of industrial enzymes using fungal cell factories. In *Grand challenges in biology and biotechnology* (pp. 179–210). Denmark: Springer Science and Business Media B.V. https://doi.org/10.1007/978-3-030-29541-7_7. Available from: http://springer.com/series/13485.

Baghban, R., Farajnia, S., Rajabibazl, M., Ghasemi, Y., Mafi, A. A., Hoseinpoor, R., Rahbarnia, L., & Aria, M. (2019). Yeast expression systems: Overview and recent advances. *Molecular Biotechnology*, *61*(5), 365–384. 1073-6085 https://doi.org/10.1007/s12033-019-00164-8.

Bahar, A. A., & Ren, D. (2013). Antimicrobial peptides. *Pharmaceuticals*, *6*(12), 1543–1575. 14248247. MDPI AG, United States. Available from: http://www.mdpi.com/1424-8247/6/12/1543/pdf. https://doi.org/10.3390/ph6121543.

Baneyx, F., & Mujacic, M. (2004). Recombinant protein folding and misfolding in Escherichia coli. *Nature Biotechnology*, *22*(11), 1399–1408. 1087-0156 https://doi.org/10.1038/nbt1029.

Bang, S. K., Kang, C. S., Han, M. D., & Bang, I. S. (2010). Expression of recombinant hybrid peptide hinnavin II/α-melanocyte-stimulating hormone in Escherichia coli: Purification and characterization. *Journal of Microbiology*, *48*(1), 24–29. 12258873 https://doi.org/10.1007/s12275-009-0317-1.

Basanta, A., Gómez-Sala, B., Sánchez, J., Diep, D. B., Herranz, C., Hernández, P. E., & Cintas, L. M. (2010). Use of the yeast pichia pastoris as an expression host for secretion of enterocin L50, a leaderless two-peptide (L50A and L50B) bacteriocin from enterococcus faecium L50▽. *Applied and Environmental Microbiology*, *76*(10), 3314–3324. 10985336 http://aem.asm.org/cgi/reprint/76/10/3314. https://doi.org/10.1128/AEM.02206-09Spain.

Bogomolovas, J., Simon, B., Sattler, M., & Stier, G. (2009). Screening of fusion partners for high yield expression and purification of bioactive viscotoxins. *Protein Expression and Purification*, *64*(1), 16–23. 10465928 https://doi.org/10.1016/j.pep.2008.10.003.

Bundy, B. C., Hunt, J. P., Jewett, M. C., Swartz, J. R., Wood, D. W., Frey, D. D., & Rao, G. (2018). Cell-free biomanufacturing. *Current Opinion in Chemical Engineering*, *22*, 177–183. 22113398. Elsevier Ltd, United States. Available from: http://www.elsevier.com/wps/find/journaldescription.cws_home/725837/description#description. https://doi.org/10.1016/j.coche.2018.10.003.

Burr, R., Stewart, E. V., & Espenshade, P. J. (2017). Coordinate regulation of yeast sterol regulatory element-binding protein (SREBP) and Mga2 transcription factors. *The Journal of Biological Chemistry*, *292*(13), 5311–5324. 00219258 https://doi.org/10.1074/jbc.m117.778209.

Burrowes, O. J., Diamond, G., & Lee, T. C. (2005). Recombinant expression of pleurocidin cDNA using the Pichia pastoris expression system. *Journal of Biomedicine and Biotechnology*, *2005*(4), 374–384. 11107251. United States. Available from: http://www.hindawi.net/access/get.aspx?journal=jbb&volume=2005&pii=S1110724305409067. https://doi.org/10.1155/JBB.2005.374.

Busso, D., Kim, R., & Kim, S. H. (2003). Expression of soluble recombinant proteins in a cell-free system using a 96-well format. *Journal of Biochemical and Biophysical Methods*, *55*(3), 233–240. 0165022X. Elsevier, United States. Available from: http://www.elsevier.com/locate/jbbm. https://doi.org/10.1016/S0165-022X(03)00049-6.

Butt, T. R., Edavettal, S. C., Hall, J. P., & Mattern, M. R. (2005). SUMO fusion technology for difficult-to-express proteins. *Protein Expression and Purification*, *43*(1), 1–9. 10465928 https://doi.org/10.1016/j.pep.2005.03.016.

Cabral, K. M. S., Almeida, M. S., Valente, A. P., Almeida, F. C. L., & Kurtenbach, E. (2003). Production of the active antifungal *Pisum sativum* defensin 1 (Psd1) in Pichia pastoris: Overcoming the inefficiency of the STE13 protease. *Protein Expression and Purification*, *31*(1), 115–122. 10465928. Academic Press Inc., Brazil. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1016/S1046-5928(03)00136-0.

Canales, J., Avila, C., & Cánovas, F. M. (2011). A maritime pine antimicrobial peptide involved in ammonium nutrition. *Plant, Cell and Environment*, *34*(9), 1443–1453. 13653040 https://doi.org/10.1111/j.1365-3040.2011.02343.x.

Cao, J., de la Fuente-Nunez, C., Ou, R. W., Torres, M. D. T., Pande, S. G., Sinskey, A. J., & Lu, T. K. (2018). Yeast-based synthetic biology platform for antimicrobial peptide production. *ACS Synthetic Biology*, *7*(3), 896–902. 2161-5063 https://doi.org/10.1021/acssynbio.7b00396.

Cao, W., Zhou, Y., Ma, Y., Luo, Q., & Wei, D. (2005). Expression and purification of antimicrobial peptide adenoregulin with C-amidated terminus in *Escherichia coli*. *Protein Expression and Purification*, *40*(2), 404–410. Academic Press Inc., China. 10465928. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1016/j.pep.2004.12.007.

Capparelli, R., Ventimiglia, I., Palumbo, D., Nicodemo, D., Salvatore, P., Amoroso, M. G., & Iannaccone, M. (2007). Expression of recombinant puroindolines for the treatment of staphylococcal skin infections (acne vulgaris). *Journal of Biotechnology*, *128*(3), 606–614. 01681656 https://doi.org/10.1016/j.jbiotec.2006.11.004.

Caschera, F., & Noireaux, V. (2014). Synthesis of 2.3 mg/ml of protein with an all Escherichia coli cell-free transcription-translation system. *Biochimie*, *99*(1), 162–168. 61831638 https://doi.org/10.1016/j.biochi.2013.11.025.

Cavicchioli, R., Ripple, W. J., Timmis, K. N., Azam, F., Bakken, L. R., Baylis, M., Behrenfeld, M. J., Boetius, A., Boyd, P. W., Classen, A. T., Crowther, T. W., Danovaro, R., Foreman, C. M., Huisman, J., Hutchins, D. A., Jansson, J. K., Karl, D. M., Koskella, B., Mark, D. B., … Webster. (2019). Scientists' warning to humanity: Microorganisms and climate change. *Nature Reviews. Microbiology*, *17*(9), 569–586. https://doi.org/10.1038/s41579-019-0222-5.

Chen, J. J., Chen, G. H., Hsu, H. C., Li, S. S., & Chen, C. S. (2004). Cloning and functional expression of a Mungbean Defensin VrD1 in Pichia pastoris. *Journal of Agricultural and Food Chemistry*, *52*(8), 2256–2261. 00218561 https://doi.org/10.1021/jf030662i.

Chen, Z., Wang, D., Cong, Y., Wang, J., Zhu, J., Yang, J., Hu, Z., Hu, X., Tan, Y., Hu, F., & Rao, X. (2011). Recombinant antimicrobial peptide hPAB-β expressed in Pichia pastoris, a potential agent active against methicillin-resistant Staphylococcus aureus. *Applied Microbiology and Biotechnology*, *89*(2), 281–291. 01757598 https://doi.org/10.1007/s00253-010-2864-0.

Chen, Y. Q., Zhang, S. Q., Li, B. C., Qiu, W., Jiao, B., Zhang, J., & Diao, Z. Y. (2008). Expression of a cytotoxic cationic antibacterial peptide in Escherichia coli using two fusion partners. *Protein Expression and Purification*, *57*(2), 303–311. 10465928 https://doi.org/10.1016/j.pep.2007.09.012.

Chen, X., Zhu, F., Cao, Y., & Qiao, S. (2009). Novel expression vector for secretion of Cecropin AD in *Bacillus subtilis* with enhanced antimicrobial activity. *Antimicrobial Agents and Chemotherapy*, *53*(9), 3683–3689. 0066-4804 https://doi.org/10.1128/aac.00251-09.

Cheng, X., Lu, W., Zhang, S., & Cao, P. (2010). Expression and purification of antimicrobial peptide CM4 by Npro fusion technology in *E. Coli*. *Amino Acids*, *39*(5), 1545–1552. 09394451 https://doi.org/10.1007/s00726-010-0625-0.

Cipakova, I., & Hostinova, E. (2005). Production of the human-beta-defensin using Saccharomyces cerevisiae as a host. *Protein and Peptide Letters*, *12*(6), 551–554. 09298665 https://doi.org/10.2174/0929866054395761.

Davydova, E. K. (2022). Protein engineering: Advances in phage display for basic science and medical research. *Biochemistry (Moscow)*, *87*, S146. 16083040. Pleiades Journals, United States. Available from: https://link.springer.com/journal/10541. https://doi.org/10.1134/S0006297922140127.

De Beer, A., & Vivier, M. A. (2008). Vv-AMP1, a ripening induced peptide from Vitis vinifera shows strong antifungal activity. *BMC Plant Biology*, *8*. https://doi.org/10.1186/1471-2229-8-75. 14712229.

Demain, A. L., & Vaishnav, P. (2009). Production of recombinant proteins by microbes and higher organisms. *Biotechnology Advances*, *27*(3), 297–306. 07349750 https://doi.org/10.1016/j.biotechadv.2009.01.008.

den Haan, R., Rose, S. H., Cripwell, R. A., Trollope, K. M., Myburgh, M. W., Viljoen-Bloom, M., & van Zyl, W. H. (2021). Heterologous production of cellulose-and starch-degrading hydrolases to expand Saccharomyces cerevisiae substrate utilization: Lessons learnt. *Biotechnology Advances*, *53*, 107859.

Deng, T., Ge, H., He, H., Liu, Y., Zhai, C., Feng, L., & Yi, L. (2017). The heterologous expression strategies of antimicrobial peptides in microbial systems. *Protein Expression and Purification*, *140*, 52–59. 10465928. Academic Press Inc., China. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1016/j.pep.2017.08.003.

Deo, S., Turton, K. L., Kainth, T., Kumar, A., & Wieden, H.-J. (2022). Strategies for improving antimicrobial peptide production. *Biotechnology Advances*, *59*, 107968. 07349750 https://doi.org/10.1016/j.biotechadv.2022.107968.

Dopp, B. J. L., Tamiev, D. D., & Reuel, N. F. (2019). Cell-free supplement mixtures: Elucidating the history and biochemical utility of additives used to support in vitro protein synthesis in *E. coli* extract. *Biotechnology Advances*, *37*(1), 246–258. https://doi.org/10.1016/j.biotechadv.2018.12.006.

Duvick, J. P., Rood, T., Rao, A. G., & Marshak, D. R. (1992). Purification and characterization of a novel antimicrobial peptide from maize (*Zea mays* L.) kernels. *The Journal of Biological Chemistry*, *267*(26), 18814–18820.

Ecker, D. M., Jones, S. D., & Levine, H. L. (2015). The therapeutic monoclonal antibody market. *MAbs*, *7*(1), 9–14. 19420870. Landes Bioscience, United States. Available from: http://www.tandfonline.com/doi/pdf/10.4161/19420862.2015.989042. https://doi.org/10.4161/19420862.2015.989042.

Egido, J. E., Toner-Bartelds, C., Costa, A. R., Brouns, S. J. J., Rooijakkers, S. H. M., Bardoel, B. W., & Haas, P.-J. (2023). Monitoring phage-induced lysis of gram-negatives in real time using a fluorescent DNA dye. *Scientific Reports*, *13*(1). https://doi.org/10.1038/s41598-023-27734-w. 2045-2322.

Endo, Y., & Sawasaki, T. (2004). High-throughput, genome-scale protein production method based on the wheat germ cell-free expression system. *Journal of Structural and Functional Genomics*, *5*(1–2), 45–57. https://doi.org/10.1023/B:JSFG.0000029208.83739.49.

Fauconnier, A. (2019). Phage therapy regulation: From night to dawn. *Viruses*, ***11***(4), 352. 1999-4915 https://doi.org/10.3390/v11040352.

Feng, X., Liu, C., Guo, J., Bi, C., Cheng, B., Li, Z., Shan, A., & Li, Z. (2010). Expression and purification of an antimicrobial peptide, bovine lactoferricin derivative LfcinB-W10 in escherichia coli. *Current Microbiology*, *60*(3), 179–184. 14320991 https://doi.org/10.1007/s00284-009-9522-8.

Feng, X., Liu, C., Guo, J., Song, X., Li, J., Xu, W., & Li, Z. (2012). Recombinant expression, purification, and antimicrobial activity of a novel hybrid antimicrobial peptide LFT33. *Applied Microbiology and Biotechnology*, *95*(5), 1191–1198. 14320614 https://doi.org/10.1007/s00253-011-3816-z.

Freudenberg, R. A., Baier, T., Einhaus, A., Wobbe, L., & Kruse, O. (2021). High cell density cultivation enables efficient and sustainable recombinant polyamine production in the microalga *Chlamydomonas reinhardtii*. *Bioresource Technology*, *323*, 124542. 09608524 https://doi.org/10.1016/j.biortech.2020.124542.

Frokjaer, S., & Otzen, D. E. (2005). Protein drug stability: A formulation challenge. *Nature Reviews. Drug Discovery*, *4*(4), 298–306. https://doi.org/10.1038/nrd1695.

Fujimoto, K., Kajino, M., & Inouye, M. (2008). Development of a series of cross-linking agents that effectively stabilize α-helical structures in various short peptides. *Chemistry- -A European Journal*, *14*(3), 857–863. 15213765 https://doi.org/10.1002/chem.200700843.

Fukuzawa, N., Tabayashi, N., Okinaka, Y., Furusawa, R., Furuta, K., Kagaya, U., & Matsumura, T. (2010). Production of biologically active Atlantic salmon interferon in transgenic potato and rice plants. *Journal of Bioscience and Bioengineering*, *110*(2), 201–207. 13891723 https://doi.org/10.1016/j.jbiosc.2010.02.005.

Gagliardo, B., Faye, A., Jaouen, M., Deschemin, J. C., Canonne-Hergaux, F., Vaulont, S., & Sari, M. A. (2008). Production of biologically active forms of recombinant hepcidin, the iron-regulatory hormone. *FEBS Journal*, *275*(15), 3793–3803. 17424658 https://doi.org/10.1111/j.1742-4658.2008.06525.x.

Gaglione, R., Pane, K., Dell'Olmo, E., Cafaro, V., Pizzo, E., Olivieri, G., Notomista, E., & Arciello, A. (2019). Cost-effective production of recombinant peptides in Escherichia coli. *New Biotechnology*, *51*, 39–48. 18764347. Elsevier B.V., Italy. Available from: http://www.elsevier.com. https://doi.org/10.1016/j.nbt.2019.02.004.

Ganz, T. (2003). The role of antimicrobial peptides in innate immunity. *Integrative and Comparative Biology*, *43*, 300–304. Society for Integrative and Comparative Biology, United States http://icb.oxfordjournals.org/. https://doi.org/10.1093/icb/43.2.300.

Garcia-Brand, A. J., Quezada, V., Gonzalez-Melo, C., Bolaños-Barbosa, A. D., Cruz, J. C., & Reyes, L. H. (2022). Novel developments on stimuli-responsive probiotic encapsulates: From smart hydrogels to nanostructured platforms. *Fermentation*, *8*(3). 23115637. MDPI, Colombia. Available from: https://www.mdpi.com/2311-5637/8/3/117/pdf. https://doi.org/10.3390/fermentation8030117.

Garvey, M. (2022). Non-mammalian eukaryotic expression systems yeast and Fungi in the production of biologics. *Journal of Fungi*, *8*(11), 1179. 2309-608X https://doi.org/10.3390/jof8111179.

Gerngross, T. U. (2004). Advances in the production of human therapeutic proteins in yeasts and filamentous fungi. *Nature Biotechnology*, *22*(11), 1409–1414. 1087-0156 https://doi.org/10.1038/nbt1028.

Gnügge, R., & Rudolf, F. (2017). Saccharomyces cerevisiae shuttle vectors. *Yeast*, *34*(5), 205–221. 0749503X https://doi.org/10.1002/yea.3228.

Gomord, V., & Faye, L. (2004). Posttranslational modification of therapeutic proteins in plants. *Current Opinion in Plant Biology*, *7*(2), 171–181. https://doi.org/10.1016/j.pbi.2004.01.015.

Gould, A., Li, Y., Majumder, S., Garcia, A. E., Carlsson, P., Shekhtman, A., & Camarero, J. A. (2012). Recombinant production of rhesus -defensin-1 (RTD-1) using a bacterial expression system. *Molecular BioSystems*, *8*(4), 1359–1365. 17422051 https://doi.org/10.1039/c2mb05451e.

Grama, S. B., Liu, Z., & Li, J. (2022). Emerging trends in genetic engineering of microalgae for commercial applications. *Marine Drugs*, *20*(5), 285. 1660-3397 https://doi.org/10.3390/md20050285.

Gueguen, Y., Herpin, A., Aumelas, A., Garnier, J., Fievet, J., Escoubas, J. M., Bulet, P., Gonzalez, M., Lelong, C., Favrel, P., & Bachère, E. (2006). Characterization of a defensin from the oyster Crassostrea gigas: Recombinant production, folding, solution structure, antimicrobial activities, and gene expression. *The Journal of Biological Chemistry*, *281*(1), 313–323. 1083351X. France. Available from: http://www.jbc.org/cgi/reprint/281/1/313.pdf. https://doi.org/10.1074/jbc.M510850200.

Gündüz Ergün, B., Hüccetoğulları, D., Öztürk, S., Çelik, E., & Çalık, P. (2019). Established and upcoming yeast expression systems. *Methods in Molecular Biology*, *1923*, 1–74. Springer Science and Business Media LLC https://doi.org/10.1007/978-1-4939-9024-5_1.

Guo, C., Huang, Y., Zheng, H., Tang, L., He, J., Xiang, L., Liu, D., & Jiang, H. (2012). Secretion and activity of antimicrobial peptide cecropin D expressed in Pichia pastoris. *Experimental and Therapeutic Medicine*, *4*(6), 1063–1068. 1792-0981 https://doi.org/10.3892/etm.2012.719.

Gutiérrez, J., Criado, R., Martín, M., Herranz, C., Cintas, L. M., & Hernández, P. E. (2005). Production of enterocin P, an antilisterial pediocin-like bacteriocin from enterococcus faecium P13, in Pichia pastons. *Antimicrobial Agents and Chemotherapy*, *49*(7), 3004–3008. https://doi.org/10.1128/AAC.49.7.3004-3008.2005.

Han, C., Fang, L., Song, S., & Min, W. (2022). Polysaccharides-based delivery system for efficient encapsulation and controlled release of food-derived active peptides. *Carbohydrate Polymers*, *291*, 119580. 01448617 https://doi.org/10.1016/j.carbpol.2022.119580.

Hancock, R. E. W., & Sahl, H. G. (2006). Antimicrobial and host-defense peptides as new anti-infective therapeutic strategies. *Nature Biotechnology*, *24*(12), 1551–1557. 15461696 https://doi.org/10.1038/nbt1267.

Hara, S., Mukae, H., Sakamoto, N., Ishimoto, H., Amenomori, M., Fujita, H., Ishimatsu, Y., Yanagihara, K., & Kohno, S. (2008). Plectasin has antibacterial activity and no affect on cell viability or IL-8 production. *Biochemical and Biophysical Research Communications*, *374*(4), 709–713. 10902104 https://doi.org/10.1016/j.bbrc.2008.07.093.

Harrison, S. J., McManus, A. M., Marcus, J. P., Goulter, K. C., Green, J. L., Nielsen, K. J., Craik, D. J., MacLean, D. J., & Manners, J. M. (1999). Purification and characterization of a plant antimicrobial peptide expressed in *Escherichia coli*. *Protein Expression and Purification*, *15*(2), 171–177. Academic Press Inc., Australia. 10465928. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1006/prep.1998.0992.

Havarstein, L. S., Diep, D. B., & Nes, I. F. (1995). A family of bacteriocin ABC transporters carry out proteolytic processing of their substrates concomitant with export. *Molecular Microbiology*, *16*(2), 229–240. 13652958 https://doi.org/10.1111/j.1365-2958.1995.tb02295.x.

Hernandez-Garcia, C. M., Bouchard, R. A., Rushton, P. J., Jones, M. L., Chen, X., Timko, M. P., & Finer, J. J. (2010). High level transgenic expression of soybean (*Glycine max*) GmERF and Gmubi gene promoters isolated by a novel promoter analysis pipeline. *BMC Plant Biology*, *10*. 14712229. United States http://www.biomedcentral.com/1471-2229/10/237. https://doi.org/10.1186/1471-2229-10-237.

Hong, I. P., Lee, S. J., Kim, Y. S., & Choi, S. G. (2007). Recombinant expression of human cathelicidin (hCAP18/LL-37) in Pichia pastoris. *Biotechnology Letters*, *29*(1), 73–78. 15736776 https://doi.org/10.1007/s10529-006-9202-8.

Huang, L., Ching, C. B., Jiang, R., & Leong, S. S. J. (2008). Production of bioactive human beta-defensin 5 and 6 in Escherichia coli by soluble fusion expression. *Protein Expression and Purification*, *61*(2), 168–174. 10465928 https://doi.org/10.1016/j.pep.2008.05.016.

Huang, L., Leong, S. S. J., & Jiang, R. (2009). Soluble fusion expression and characterization of bioactive human beta-defensin 26 and 27. *Applied Microbiology and Biotechnology*, *84*(2), 301–308. 0175–7598 https://doi.org/10.1007/s00253-009-1982-z.

Huang, L., Wang, J., Zhong, Z., Peng, L., Chen, H., Xu, Z., & Cen, P. (2006). Production of bioactive human β-defensin-3 in Escherichia coli by soluble fusion expression. *Biotechnology Letters*, *28*(9), 627–632. 15736776 https://doi.org/10.1007/s10529-006-0024-5.

Imjongjirak, C., Amparyup, P., Tassanakajon, A., & Sittipraneed, S. (2009). Molecular cloning and characterization of crustin from mud crab Scylla paramamosain. *Molecular Biology Reports*, *36*(5), 841–850. 15734978 https://doi.org/10.1007/s11033-008-9253-0.

Ingham, A. B., & Moore, R. J. (2007). Recombinant production of antimicrobial peptides in heterologous microbial systems. *Biotechnology and Applied Biochemistry*, *47*(1), 1–9. 08854513 https://doi.org/10.1042/BA20060207.

Jang, S. A., Sung, B. H., Cho, J. H., & Kim, S. C. (2009). Direct expression of antimicrobial peptides in an intact form by a translationally coupled two-Cistron expression system. *Applied and Environmental Microbiology*, *75*(12), 3980–3986. 0099-2240 https://doi.org/10.1128/aem.02753-08.

Jenkins, N., Murphy, L., & Tyther, R. (2008). Post-translational modifications of recombinant proteins: Significance for biopharmaceuticals. *Molecular Biotechnology*, *39*, 113–118. 10736085. Ireland https://doi.org/10.1007/s12033-008-9049-4.

Jermutus, L., Ryabova, L. A., & Plückthun, A. (1998). Recent advances in producing and selecting functional proteins by using cell-free translation. *Current Opinion in Biotechnology*, *9*(5), 534–548. 09581669 https://doi.org/10.1016/s0958-1669(98)80042-6.

Jewett, M. C., Calhoun, K. A., Voloshin, A., Wuu, J. J., & Swartz, J. R. (2008). An integrated cell-free metabolic platform for protein production and synthetic biology. *Molecular Systems Biology*, *4*. https://doi.org/10.1038/msb.2008.57. 17444292.

Jewett, M. C., & Swartz, J. R. (2004a). Mimicking the Escherichia coli cytoplasmic environment activates long-lived and efficient cell-free protein synthesis. *Biotechnology and Bioengineering*, *86*(1), 19–26. https://doi.org/10.1002/bit.20026.

Jewett, M. C., & Swartz, J. R. (2004b). Substrate replenishment extends protein synthesis with an in vitro translation system designed to mimic the cytoplasm. *Biotechnology and Bioengineering*, *87*(4), 465–471. 00063592. John Wiley and Sons Inc., United States. Available from: http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-0290. https://doi.org/10.1002/bit.20139.

Jin, F., Xu, X., Zhang, W., & Gu, D. (2006). Expression and characterization of a housefly cecropin gene in the methylotrophic yeast, Pichia pastoris. *Protein Expression and Purification*, *49*(1), 39–46. 10465928 https://doi.org/10.1016/j.pep.2006.03.008.

Jing, X. L., Luo, X. G., Tian, W. J., Lv, L. H., Jiang, Y., Wang, N., & Zhang, T. C. (2010). High-level expression of the antimicrobial peptide plectasin in escherichia coli. *Current Microbiology*, *61*(3), 197–202. 14320991 https://doi.org/10.1007/s00284-010-9596-3.

Kane, P. M., Yamashiro, C. T., Wolczyk, D. F., Neff, N., Goebl, M., & Stevens, T. H. (1990). Protein splicing converts the yeast TFP1 gene product to the 69-kD subunit of the vacuolar H+-adenosine triphosphatase. *Science*, *250*(4981), 651–657. 00368075 https://doi.org/10.1126/science.2146742.

Kang, Z., Huang, H., Zhang, Y., Guocheng, D., & Chen, J. (2017). Recent advances of molecular toolbox construction expand Pichia pastoris in synthetic biology applications. *World Journal of Microbiology and Biotechnology*, *33*(1). https://doi.org/10.1007/s11274-016-2185-2. 0959-3993.

Kang, C. S., Son, S. Y., & Bang, I. S. (2008). Biologically active and C-amidated hinnavinII-38-Asn produced from a Trx fusion construct in Escherichia coli. *Journal of Microbiology*, *46*(6), 656–661. 12258873 https://doi.org/10.1007/s12275-008-0214-z.

Kant, P., Liu, W. Z., & Pauls, K. P. (2009). PDC1, a corn defensin peptide expressed in Escherichia coli and Pichia pastoris inhibits growth of fusarium graminearum. *Peptides*, *30*(9), 1593–1599. 01969781 https://doi.org/10.1016/j.peptides.2009.05.024.

Karagiosis, S. A., & Baker, S. E. (2012). *Fungal cell factories food and industrial bioproducts and bioprocessing* (pp. 205–219). United States: Wiley-Blackwell. Available from: http://onlinelibrary.wiley.com/book/10.1002/9781119946083. https://doi.org/10.1002/9781119946083.ch8.

Karbalaei, M., Rezaee, S. A., & Farsiani, H. (2020). Pichia pastoris: A highly successful expression system for optimal synthesis of heterologous proteins. *Journal of Cellular Physiology*, *235*(9), 5867–5881. 0021-9541 https://doi.org/10.1002/jcp.29583.

Khnouf, R., Beebe, D. J., & Fan, Z. H. (2009). Cell-free protein expression in a microchannel array with passive pumping. *Lab on a Chip*, ***9***(1), 56–61. 14730189. Royal Society of Chemistry, United States. Available from: http://pubs.rsc.org/en/journals/journal/lc. https://doi.org/10.1039/b808034h.

Kim, H. K., Chun, D. S., Kim, J. S., Yun, C. H., Lee, J. H., Hong, S. K., & Kang, D. K. (2006). Expression of the cationic antimicrobial peptide lactoferricin fused with the anionic peptide in Escherichia coli. *Applied Microbiology and Biotechnology*, *72*(2), 330–338. 01757598 https://doi.org/10.1007/s00253-005-0266-5.

Kim, J. M., Jang, S. A., Yu, B. J., Sung, B. H., Cho, J. H., & Kim, S. C. (2008). High-level expression of an antimicrobial peptide histonin as a natural form by multimerization and furin-mediated cleavage. *Applied Microbiology and Biotechnology*, *78*(1), 123–130. 01757598 https://doi.org/10.1007/s00253-007-1273-5.

Kim, S.j., Quan, R., Lee, S. J., Lee, H. K., & Choi, J. K. (2009). Antibacterial activity of recombinant hCAP18/LL37 protein secreted from Pichia pastoris. *Journal of Microbiology*, *47*(3), 358–362. 12258873 https://doi.org/10.1007/s12275-009-0131-9.

Kim, H., Yoo, S. J., & Kang, H. A. (2015). Yeast synthetic biology for the production of recombinant therapeutic proteins. *FEMS Yeast Research*, *15*(1). 15671364. Oxford University Press, South Korea. Available from: http://femsyr.oxfordjournals.org/. https://doi.org/10.1111/1567-1364.12195.

Lai, Y. P., Peng, Y. F., Zuo, Y., Li, J., Huang, J., Wang, L. F., & Wu, Z. R. (2005). Functional and structural characterization of recombinant dermcidin-1L, a human antimicrobial peptide. *Biochemical and Biophysical Research Communications*, *328*(1), 243–250. 0006291X https://doi.org/10.1016/j.bbrc.2004.12.143.

LaVallie, E. R., DiBlasio, E. A., Kovacic, S., Grant, K. L., Schendel, P. F., & McCoy, J. M. (1993). A thioredoxin gene fusion expression system that circumvents inclusion body formation in the *E. Coli* cytoplasm. *Bio/Technology*, *11*(2), 187–193. 0733222X https://doi.org/10.1038/nbt0293-187.

LaVallie, E. R., Lu, Z., Diblasio-Smith, E. A., Collins-Racie, L. A., & Mccoy, J. M. (2000). Thioredoxin as a fusion partner for production of soluble recombinant proteins in Escherichia coli. *Methods in Enzymology*, *326*, 322–340. 00766879. Academic Press Inc., undefined. Available from: https://www.sciencedirect.com/bookseries/methods-in-enzymology. https://doi.org/10.1016/s0076-6879(00)26063-1.

Lee, J. W., Chan, C. T. Y., Slomovic, S., & Collins, J. J. (2018). Next-generation biocontainment systems for engineered organisms. *Nature Chemical Biology*, *14*(6), 530–537. Nature Publishing Group, United States. 15524469. Available from: http://www.nature.com/nchembio. https://doi.org/10.1038/s41589-018-0056-x.

Lee, J. H., Kim, J. H., Hwang, S. W., Lee, W. J., Yoon, H. K., Lee, H. S., & Hong, S. S. (2000). High-level expression of antimicrobial peptide mediated by a fusion partner reinforcing formation of inclusion bodies. *Biochemical and Biophysical Research Communications*, *277*(3), 575–580. 0006291X. Academic Press Inc., South Korea. Available from: http://www.sciencedirect.com/science/journal/0006291X. https://doi.org/10.1006/bbrc.2000.3712.

Lee, K. H., Kwon, Y. C., Yoo, S. J., & Kim, D. M. (2010). Ribosomal synthesis and in situ isolation of peptide molecules in a cell-free translation system. *Protein Expression and Purification*, *71*(1), 16–20. 10465928 https://doi.org/10.1016/j.pep.2010.01.016.

Lee, J. H., Minn, I., Park, C. B., & Kim, S. C. (1998a). Acidic peptide-mediated expression of the antimicrobial peptide Buforin II as tandem repeats in Escherichia coli. *Protein Expression and Purification*, *12*(1), 53–60. 10465928 https://doi.org/10.1006/prep.1997.0814.

Lee, J. H., Minn, I., Park, C. B., & Kim, S. C. (1998b). Acidic peptide-mediated expression of the antimicrobial peptide buforin II as tandem repeats in Escherichia coli. *Protein Expression and Purification*, *12*(1), 53–60. 10465928. Academic Press Inc., South Korea. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1006/prep.1997.0814.

Lee, S. J., Park, I. S., Han, Y. H., Kim, Y. O., & Reeves, P. R. (2008). Soluble expression of recombinant olive flounder hepcidin I using a novel secretion enhancer. *Molecules and Cells*, *26*(2), 140–145. 10168478. South Korea. Available from: http://molcells.org/home/journal/include/downloadPdf.asp?articleuid={0D7B45C2-19B4-4E11-BE3D-8D1B5B31CB52}.

Lesley, S. A., Brow, M. A., & Burgess, R. R. (1991). Use of in vitro protein synthesis from polymerase chain reaction-generated templates to study interaction of Escherichia coli transcription factors with core RNA polymerase and for epitope mapping of monoclonal antibodies. *The Journal of Biological Chemistry*, *266*(4), 2632–2638. 00219258 https://doi.org/10.1016/s0021-9258(18)52291-2.

Li, Y. (2009). Carrier proteins for fusion expression of antimicrobial peptides in *Escherichia coli*. *Biotechnology and Applied Biochemistry*, *54*(1), 1–9. 0885-4513 https://doi.org/10.1042/ba20090087.

Li, Y. (2011a). Recombinant production of antimicrobial peptides in Escherichia coli: A review. *Protein Expression and Purification*, *80*(2), 260–267. 10465928 https://doi.org/10.1016/j.pep.2011.08.001.

Li, Y. (2011b). Self-cleaving fusion tags for recombinant protein production. *Biotechnology Letters*, *33*(5), 869–881. 15736776 https://doi.org/10.1007/s10529-011-0533-8.

Li, Y. (2013). Production of human antimicrobial peptide LL-37 in Escherichia coli using a thioredoxin-SUMO dual fusion system. *Protein Expression and Purification*, *87*(2), 72–78. https://doi.org/10.1016/j.pep.2012.10.008.

Li, P., Anumanthan, A., Gao, X.-G., Ilangovan, K., Suzara, V. V., Düzgüneş, N., & Renugopalakrishnan, V. (2007). Expression of recombinant proteins in pichia pastoris. *Applied Biochemistry and Biotechnology*, *142*(2), 105–124. 0273-2289 https://doi.org/10.1007/s12010-007-0003-x.

Li, Y., & Chen, Z. (2008). RAPD: A database of recombinantly-produced antimicrobial peptides. *FEMS Microbiology Letters*, *289*(2), 126–129. 15746968 https://doi.org/10.1111/j.1574-6968.2008.01357.x.

Li, C., Haug, T., Styrvold, O. B., Jørgensen, T. O., & Stensvåg, K. (2008). Strongylocins, novel antimicrobial peptides from the green sea urchin, Strongylocentrotus droebachiensis. *Developmental and Comparative Immunology*, *32*(12), 1430–1440. 0145305X https://doi.org/10.1016/j.dci.2008.06.013.

Li, Y., Li, X., Li, H., Lockridge, O., & Wang, G. (2007). A novel method for purifying recombinant human host defense cathelicidin LL-37 by utilizing its inherent property of aggregation. *Protein Expression and Purification*, *54*(1), 157–165. 10465928 https://doi.org/10.1016/j.pep.2007.02.003.

Li, Y., Li, X., & Wang, G. (2006). Cloning, expression, isotope labeling, and purification of human antimicrobial peptide LL-37 in Escherichia coli for NMR studies. *Protein Expression and Purification*, *47*(2), 498–505. 10465928 https://doi.org/10.1016/j.pep.2005.10.022.

Li, Q., Lu, J., Zhang, G., Liu, S., Zhou, J., Du, G., & Chen, J. (2022). Recent advances in the development of Aspergillus for protein production. *Bioresource Technology*, *348*. 18732976. Elsevier Ltd, China. Available from: http://www.elsevier.com/locate/biortech. https://doi.org/10.1016/j.biortech.2022.126768.

Li, L., Wang, J. X., Zhao, X. F., Kang, C. J., Liu, N., Xiang, J. H., Li, F. H., Sueda, S., & Kondo, H. (2005). High level expression, purification, and characterization of the shrimp antimicrobial peptide, Ch-penaeidin, in Pichia pastoris. *Protein Expression and Purification*, *39*(2), 144–151. 10465928 https://doi.org/10.1016/j.pep.2004.09.006.

Li, B. C., Zhang, S. Q., Dan, W. B., Chen, Y. Q., & Cao, P. (2007). Expression in Escherichia coli and purification of bioactive antibacterial peptide ABP-CM4 from the Chinese silk worm, *Bombyx mori*. *Biotechnology Letters*, *29*(7), 1031–1036. 15736776 https://doi.org/10.1007/s10529-007-9351-4.

Li, J. F., Zhang, J., Song, R., Zhang, J. X., Shen, Y., & Zhang, S. Q. (2009). Production of a cytotoxic cationic antibacterial peptide in Escherichia coli using SUMO fusion partner. *Applied Microbiology and Biotechnology*, *84*(2), 383–388. 01757598 https://doi.org/10.1007/s00253-009-2109-2.

Li, J. F., Zhang, J., Zhang, Z., Kang, C. T., & Zhang, S. Q. (2011). SUMO mediating fusion expression of antimicrobial peptide CM4 from two joined genes in escherichia coli. *Current Microbiology*, *62*(1), 296–300. 14320991 https://doi.org/10.1007/s00284-010-9705-3.

Li, J. F., Zhang, J., Zhang, Z., Ma, H. W., Zhang, J. X., & Zhang, S. Q. (2010). Production of bioactive human beta-defensin-4 in *Escherichia coli* using SUMO fusion partner. *Protein Journal*, *29*(5), 314–319. 18758355. Springer Science and Business Media, LLC, China. Available from: http://www.kluweronline.com/issn/1572-3887. https://doi.org/10.1007/s10930-010-9254-4.

Liang, Q., Cao, L., Zhu, C., Kong, Q., Sun, H., Zhang, F., Mou, H., & Liu, Z. (2022). Characterization of recombinant antimicrobial peptide BMGlv2 heterologously expressed in trichoderma reesei. *International Journal of Molecular Sciences*, *23*(18), 10291. 1422-0067 https://doi.org/10.3390/ijms231810291.

Liang, Y., Wang, J. X., Zhao, X. F., Du, X. J., & Xue, J. F. (2006). Molecular cloning and characterization of cecropin from the housefly (Musca domestica), and its expression in Escherichia coli. *Developmental and Comparative Immunology*, *30*(3), 249–257. 0145305X https://doi.org/10.1016/j.dci.2005.04.005.

Lin, D. M., Koskella, B., & Lin, H. C. (2017). Phage therapy: An alternative to antibiotics in the age of multi-drug resistance. *World Journal of Gastrointestinal Pharmacology and Therapeutics*, *8*(3), 162. 2150-5349 https://doi.org/10.4292/wjgpt.v8.i3.162.

Liu, Q., Song, L., Peng, Q., Zhu, Q., Shi, X., Mingqiang, X., Wang, Q., Zhang, Y., & Cai, M. (2022). A programmable high-expression yeast platform responsive to user-defined signals. *Science Advances*, *8*(6). https://doi.org/10.1126/sciadv.abl5166. 2375-2548.

López-García, B., Moreno, A. B., Segundo, B. S., De los Ríos, V., Manning, J. M., Gavilanes, J. G., & Martínez-del-Pozo, Á. (2010). Production of the biotechnologically relevant AFP from aspergillus giganteus in the yeast Pichia pastoris. *Protein Expression and Purification*, *70*(2), 206–210. 10465928 https://doi.org/10.1016/j.pep.2009.11.002.

Lu, T. K., De La Fuente-Nunez, C., Torres, M. D. T., Cao, J., & Franco, O. L. (2021). Synthetic biology and computer-based frameworks for antimicrobial peptide discovery. *ACS Nano*, *15*(2), 2143–2164. 1936086X. American Chemical Society, United States. Available from: http://pubs.acs.org/journal/ancac3. https://doi.org/10.1021/acsnano.0c09509.

Lübeck, M., & Lübeck, P. S. (2022). Fungal cell factories for efficient and sustainable production of proteins and peptides. *Microorganisms.*, *10*(4), 753. 2076-2607. Available from: https://doi.org/10.3390/microorganisms10040753.

Luo, H., Chen, S., Ren, F., Guo, H., Lin, S., & Xu, W. (2007). In vitro reconstitution of antimicrobial pathogen activity by expressed recombinant bovine lactoferrin N-terminal peptide in Escherichia coli. *The Journal of Dairy Research*, *74*(2), 233–238. 0022-0299 https://doi.org/10.1017/s0022029907002531.

Ma, D.y., Liu, S.w., Han, Z.x., Li, Y.j., & Shan, A.s. (2008). Expression and characterization of recombinant gallinacin-9 and gallinacin-8 in Escherichia coli. *Protein Expression and Purification*, *58*(2), 284–291. 10465928 https://doi.org/10.1016/j.pep.2007.11.017.

Malakhov, M. P., Mattern, M. R., Malakhova, O. A., Drinker, M., Weeks, S. D., & Butt, T. R. (2004). SUMO fusions and SUMO-specific protease for efficient expression and purification of proteins. *Journal of Structural and Functional Genomics*, *5*(1–2), 75–86. https://doi.org/10.1023/B:JSFG.0000029237.70316.52.

Malla, A., Rosales-Mendoza, S., Phoolcharoen, W., & Vimolmangkang, S. (2021). Efficient transient expression of recombinant proteins using DNA viral vectors in freshwater microalgal species. *Frontiers in Plant Science*, *12*. https://doi.org/10.3389/fpls.2021.650820. 1664-462X.

Markham, K. A., & Alper, H. S. (2018). Synthetic biology expands the industrial potential of Yarrowia lipolytica. *Trends in Biotechnology*, *36*(10), 1085–1095. 01677799 https://doi.org/10.1016/j.tibtech.2018.05.004.

Martemyanov, K. A., Spirin, A. S., & Gudkov, A. T. (1997). Direct expression of PCR products in a cell-free transcription/translation system: Synthesis of antibacterial peptide cecropin. *FEBS Letters*, *414*(2), 268–270. 00145793 https://doi.org/10.1016/S0014-5793(97)01011-9.

Martínez-Ruiz, A., Del Pozo, A. M., Lacadena, J., Mancheño, J. M., Oñaderra, M., López-Otín, C., & Gavilanes, J. G. (1998). Secretion of recombinant pro- and mature fungal α-sarcin ribotoxin by the methylotrophic yeast Pichia pastoris: The Lys-Arg motif is required for maturation. *Protein Expression and Purification*, *12*(3), 315–322. 10465928. Academic Press Inc., Spain. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1006/prep.1997.0846.

Mattanovich, D., Branduardi, P., Dato, L., Gasser, B., Sauer, M., & Porro, D. (2011). Recombinant protein production in yeasts. *Methods in Molecular Biology*, *824*, 329–358. Springer Science and Business Media LLC https://doi.org/10.1007/978-1-61779-433-9_17.

Mayeda, A., Zahler, A. M., Krainer, A. R., & Roth, M. B. (1992). Two members of a conserved family of nuclear phosphoproteins are involved in pre-mRNA splicing. *Proceedings of the National Academy of Sciences of the United States of America*, *89*(4), 1301–1304.

McClements, D. J. (2018). Encapsulation, protection, and delivery of bioactive proteins and peptides using nanoparticle and microparticle systems: A review. *Advances in Colloid and Interface Science*, *253*, 1–22. 00018686 https://doi.org/10.1016/j.cis.2018.02.002.

Mehrnejad, F., Naderi-Manesh, H., Ranjbar, B., Maroufi, B., Asoodeh, A., & Doustdar, F. (2008). PCR-based gene synthesis, molecular cloning, high level expression, purification, and characterization of novel antimicrobial peptide, Brevinin-2R, in Escherichia coli. *Applied Biochemistry and Biotechnology*, *149*(2), 109–118. 02732289 https://doi.org/10.1007/s12010-007-8024-z.

Melinek, B., Colant, C., Stamatis, C., Lennon, C., Farid, S. S., Polizzi, K., … Bracewell, D. G. (2020). Toward a roadmap for cell-free synthesis in bioprocessing. *BioProcess International*, *18*(9), 40–52.

Merk, H., Meschkat, D., & Stiege, W. (2003). *Expression-PCR: From gene pools to purified proteins within 1 day* (pp. 15–23). Springer Science and Business Media LLC. https://doi.org/10.1007/978-3-642-59337-6_3.

Metlitskaia, L., Cabralda, J. E., Suleman, D., Kerry, C., Brinkman, J., Bartfeld, D., & Guarna, M. M. (2004). Recombinant antimicrobial peptides efficiently produced using novel cloning and purification processes. *Biotechnology and Applied Biochemistry*, *39*(3), 339–345. 08854513. Portland Press Ltd, Canada. Available from: http://iubmb.onlinelibrary.wiley.com/hub/journal/10.1002/(ISSN)1470-8744/. https://doi.org/10.1042/ba20030100.

Michel-Reydellet, N., Calhoun, K., & Swartz, J. (2004). Amino acid stabilization for cell-free protein synthesis by modification of the Escherichia coli genome. *Metabolic Engineering*, *6*(3), 197–203. https://doi.org/10.1016/j.ymben.2004.01.003.

Miyazaki-Imamura, C., Oohira, K., Kitagawa, R., Nakano, H., Yamane, T., & Takahashi, H. (2003). Improvement of H2O2 stability of manganese peroxidase by combinatorial mutagenesis and high-throughput screening using in vitro expression with protein disulfide isomerase. *Protein Engineering, Design & Selection*, *16*(6), 423–428. 1741-0126 https://doi.org/10.1093/protein/gzg054.

Moon, J. Y., Henzler-Wildman, K. A., & Ramamoorthy, A. (2006). Expression and purification of a recombinant LL-37 from *Escherichia coli*. *Biochimica et Biophysica Acta - Biomembranes*, *1758*(9), 1351–1358. 00052736 https://doi.org/10.1016/j.bbamem.2006.02.003.

Morassutti, C., De Amicis, F., Bandiera, A., & Marchetti, S. (2005). Expression of SMAP-29 cathelicidin-like peptide in bacterial cells by intein-mediated system. *Protein Expression and Purification*, *39*(2), 160–168. 10465928 https://doi.org/10.1016/j.pep.2004.11.006.

Moreno Ayala, M. A., Gottardo, M. F., Zuccato, C. F., Pidre, M. L., Candia, A. J. N., Asad, A. S., Imsen, M., Romanowski, V., Creton, A., Larrain, M. I., Seilicovich, A., & Candolfi, M. (2020). Humanin promotes tumor progression in experimental triple negative breast cancer. *Scientific Reports*, *10*(1). https://doi.org/10.1038/s41598-020-65381-7.

Morin, K. M., Arcidiacono, S., Beckwitt, R., & Mello, C. M. (2006). Recombinant expression of indolicidin concatamers in Escherichia coli. *Applied Microbiology and Biotechnology*, *70*(6), 698–704. 01757598 https://doi.org/10.1007/s00253-005-0132-5.

Murray, C. J., & Baliga, R. (2013). Cell-free translation of peptides and proteins: From high throughput screening to clinical production. *Current Opinion in Chemical Biology*, *17*(3), 420–426. 18790402 https://doi.org/10.1016/j.cbpa.2013.02.014.

Mygind, P. H., Fischer, R. L., Schnorr, K. M., Hansen, M. T., Sönksen, C. P., Ludvigsen, S., Raventós, D., Buskov, S., Christensen, B., De Maria, L., Taboureau, O., Yaver, D., Elvig-Jørgensen, S. G., Sørensen, M. V., Christensen, B. E., Kjærulff, S., Frimodt-Moller, N., Lehrer, R. I., Zasloff, M., & Kristensen, H.-H. (2005). Plectasin is a peptide antibiotic with therapeutic potential from a saprophytic fungus. *Nature*, *437*(7061), 975–980. 0028-0836 https://doi.org/10.1038/nature04051.

Niimi, T. (2011). Recombinant protein production in the eukaryotic protozoan parasite Leishmania tarentolae: A review. *Methods in Molecular Biology*, *824*, 307–315. Springer Science and Business Media LLC https://doi.org/10.1007/978-1-61779-433-9_15.

Ohuchi, S., Nakano, H., & Yamane, T. (1998). In vitro method for the generation of protein libraries using PCR amplification of a single DNA molecule and coupled transcription/translation. *Nucleic Acids Research*, *26*(19), 4339–4346. 03051048 https://doi.org/10.1093/nar/26.19.4339.

Paloheimo, M., Mäntylä, A., Kallio, J., & Suominen, P. (2003). High-yield production of a bacterial xylanase in the filamentous fungus Trichoderma reesei requires a carrier polypeptide with an intact domain structure. *Applied and Environmental Microbiology*, *69*(12), 7073–7082. 0099-2240 https://doi.org/10.1128/aem.69.12.7073-7082.2003.

Parachin, N. S., Mulder, K. C., Viana, A. A. B., Dias, S. C., & Franco, O. L. (2012). Expression systems for heterologous production of antimicrobial peptides. *Peptides*, *38*(2), 446–456. 01969781 https://doi.org/10.1016/j.peptides.2012.09.020.

Patra, P., Das, M., Kundu, P., & Ghosh, A. (2021). Recent advances in systems and synthetic biology approaches for developing novel cell-factories in non-conventional yeasts. *Biotechnology Advances*, *47*, 107695. 07349750 https://doi.org/10.1016/j.biotechadv.2021.107695.

Peng, H., Liu, H. P., Chen, B., Hao, H., & Wang, K. J. (2012). Optimized production of scygonadin in Pichia pastoris and analysis of its antimicrobial and antiviral activities. *Protein Expression and Purification*, *82*(1), 37–44. 10465928 https://doi.org/10.1016/j.pep.2011.11.008.

Peng, H., Yang, M., Huang, W. S., Ding, J., Qu, H. D., Cai, J. J., Zhang, N., & Wang, K. J. (2010). Soluble expression and purification of a crab antimicrobial peptide scygonadin in different expression plasmids and analysis of its antimicrobial activity. *Protein Expression and Purification*, *70*(1), 109–115. 10465928 https://doi.org/10.1016/j.pep.2009.09.008.

Perler, F. B., Davis, E. O., Dean, G. E., Gimble, F. S., Jack, W. E., Neff, N., Noren, C. J., Thorner, J., & Belfort, M. (1994). Protein splicing elements: Inteins and exteins - A definition of terms and recommended nomenclature. *Nucleic Acids Research*, *22*(7), 1125–1127. 03051048 https://doi.org/10.1093/nar/22.7.1125.

Pham, J. V., Yilma, M. A., Feliz, A., Majid, M. T., Maffetone, N., Walker, J. R., Kim, E., Cho, H. J., Reynolds, J. M., Song, M. C., Park, S. R., & Yoon, Y. J. (2019). A review of the microbial production of bioactive natural products and biologics. *Frontiers in Microbiology*, *10*. https://doi.org/10.3389/fmicb.2019.01404. 1664-302X.

Piers, K. L., Brown, M. H., & Hancock, R. E. W. (1993). Recombinant DNA procedures for producing small antimicrobial cationic peptides in bacteria. *Gene*, *134*(1), 7–13. 03781119 https://doi.org/10.1016/0378-1119(93)90168-3.

Prielhofer, R., Barrero, J. J., Steuer, S., Gassler, T., Zahrl, R., Baumann, K., Sauer, M., Mattanovich, D., Gasser, B., & Marx, H. (2017). GoldenPiCS: A Golden Gate-derived modular cloning system for applied synthetic biology in the yeast Pichia pastoris. *BMC Systems Biology*, *11*(1). https://doi.org/10.1186/s12918-017-0492-3. 1752-0509.

Qian, Y., Fan, C., Hu, T., Yang, Y., Yang, S., Gong, Y., et al. (2000). Molecular cloning and functional expression of α-bungarotoxin (V31) from Chinese continental banded krait. *Zoological Research*, *21*, 41–47.

Rademacher, T., Sack, M., Blessing, D., Fischer, R., Holland, T., & Buyel, J. (2019). Plant cell packs: A scalable platform for recombinant protein production and metabolic engineering. *Plant Biotechnology Journal*, *17*(8), 1560–1566. 14677652. Blackwell Publishing Ltd, Germany. Available from: http://onlinelibrary.wiley.com/journal/10.1111/(ISSN)1467-7652. https://doi.org/10.1111/pbi.13081.

Rahman, M. M., Hosano, N., & Hosano, H. (2022). Recovering microalgal bioresources: A review of cell disruption methods and extraction technologies. *Molecules*, *27*(9), 2786. 1420-3049 https://doi.org/10.3390/molecules27092786.

Ramírez, L. A., Pérez, E. L., Díaz, C. G., Luengas, D. A. C., Ratkovich, N., & Reyes, L. H. (2020). CFD and experimental characterization of a bioreactor: Analysis via power curve, flow patterns and k L a. *Processes*, *8*(7), 878. 2227-9717 https://doi.org/10.3390/pr8070878.

Rao, X., Hu, J., Li, S., Jin, X., Zhang, C., Cong, Y., Hu, X., Tan, Y., Huang, J., Chen, Z., Zhu, J., & Hu, F. (2005). Design and expression of peptide antibiotic hPAB-β as tandem multimers in *Escherichia coli*. *Peptides*, *26*(5), 721–729. 01969781. Elsevier Inc., China. Available from: http://www.elsevier.com/locate/peptides. https://doi.org/10.1016/j.peptides.2004.12.016.

Rao, X. C., Li, S., Hu, J. C., Jin, X. L., Hu, X. M., Huang, J. J., Chen, Z. J., Zhu, J. M., & Hu, F. Q. (2004). A novel carrier molecule for high-level expression of peptide antibiotics in Escherichia coli. *Protein Expression and Purification*, *36*(1), 11–18. https://doi.org/10.1016/j.pep.2004.01.020.

Reyes, L. H., Cardona, C., Pimentel, L., Rodríguez-López, A., & Alméciga-Díaz, C. J. (2017). Improvement in the production of the human recombinant enzyme N-acetylgalactosamine-6-sulfatase (rhGALNS) in Escherichia coli using synthetic biology approaches. *Scientific Reports*, *7*(1). 20452322. Nature Publishing Group, Colombia. Available from: http://www.nature.com/srep/index.html. https://doi.org/10.1038/s41598-017-06367-w.

Rungpragayphan, S., Kawarasaki, Y., Imaeda, T., Kohda, K., Nakano, H., & Yamane, T. (2002). High-throughput, cloning-independent protein library construction by combining single-molecule DNA amplification with in vitro expression. *Journal of Molecular Biology*, *318*(2), 395–405. 00222836. Academic Press, Japan. Available from: https://www.journals.elsevier.com/journal-of-molecular-biology. https://doi.org/10.1016/S0022-2836(02)00094-3.

Sakekar, A. A., Gaikwad, S. R., & Punekar, N. S. (2021). Protein expression and secretion by filamentous fungi. *Journal of Biosciences*, *46*(1). https://doi.org/10.1007/s12038-020-00120-8. 0250-5991.

Salehi, A. S. M., Smith, M. T., Bennett, A. M., Williams, J. B., Pitt, W. G., & Bundy, B. C. (2016). Cell-free protein synthesis of a cytotoxic cancer therapeutic: Onconase production and a just-add-water cell-free system. *Biotechnology Journal*, *11*(2), 274–281. Wiley-VCH Verlag, United States. 18607314. Available from: http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1860-7314. https://doi.org/10.1002/biot.201500237.

Sang, M., Wei, H., Zhang, J., Wei, Z., Wu, X., Chen, Y., & Zhuge, Q. (2017). Expression and characterization of the antimicrobial peptide ABP-dHC-cecropin A in the methylotrophic yeast Pichia pastoris. *Protein Expression and Purification*, *140*, 44–51. 10465928. Academic Press Inc., China. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1016/j.pep.2017.08.001.

Sawasaki, T., Ogasawara, T., Morishita, R., & Endo, Y. (2002). A cell-free protein synthesis system for high-throughput proteomics. *Proceedings of the National Academy of Sciences of the United States of America*, *99*(23), 14652–14657. https://doi.org/10.1073/pnas.232580399.

Schoeman, H., Vivier, M. A., Du Toit, M., Dicks, L. M. T., & Pretorius, I. S. (1999). The development of bactericidal yeast strains by expressing the *Pediococcus acidilactici* pediocin gene (pedA) in *Saccharomyces cerevisiae*. *Yeast*, *15*(8), 647–656. 0749503X https://doi.org/10.1002/(SICI)1097-0061(19990615)15:8%3c647::AID-YEA409%3e3.0.CO;2-5.

Shao, J. H., Wang, Y. Q., Wu, X. Y., Jiang, R., Zhang, R., Wu, C. F., & Zhang, J. H. (2008). Cloning, expression, and pharmacological activity of BmK AS, an active peptide from scorpion Buthus martensii Karsch. *Biotechnology Letters*, *30*(1), 23–29. 15736776 https://doi.org/10.1007/s10529-007-9499-y.

Sheibani, N. (1999). Prokaryotic gene fusion expression systems and their use in structural and functional studies of proteins. *Preparative Biochemistry & Biotechnology*, *29*(1), 77–90. 10826068. Marcel Dekker Inc., United States. Available from: http://www.tandf.co.uk/journals/titles/10826068.asp. https://doi.org/10.1080/10826069908544695.

Shlyapnikov, Y. M., Andreev, Y. A., Kozlov, S. A., Vassilevski, A. A., & Grishin, E. V. (2008). Bacterial production of latarcin 2a, a potent antimicrobial peptide from spider venom. *Protein Expression and Purification*, *60*(1), 89–95. 10465928 https://doi.org/10.1016/j.pep.2008.03.011.

Shrestha, P., Holland, T. M., & Bundy, B. C. (2012). Streamlined extract preparation for *Escherichia coli*-based cell-free protein synthesis by sonication or bead vortex mixing. *BioTechniques*, *53*(3), 163–174. 19409818. United States. Available from: http://www.biotechniques.com/multimedia/archive/00184/BTN_A_0000113924_O_184273a.pdf. https://doi.org/10.2144/0000113924.

Shrivastava, A., Pal, M., & Sharma, R. K. (2023). Pichia as yeast cell factory for production of industrially important bio-products: Current trends, challenges, and future prospects. *Journal of Bioresources and Bioproducts*. https://doi.org/10.1016/j.jobab.2023.01.007. 23699698.

Si, L. G., Liu, X. C., Lu, Y. Y., Wang, G. Y., & Li, W. M. (2007). Soluble expression of active human β-defensin-3 in Escherichia coli and its effects on the growth of host cells. *Chinese Medical Journal*, *120*(8), 708–713. 03666999. Chinese Medical Association, China. Available from: http://www.cmj.org/Periodical/PDF/200742033543410.pdf. https://doi.org/10.1097/00029330-200704020-00018.

Skosyrev, V. S., Kulesskiy, E. A., Yakhnin, A. V., Temirov, Y. V., & Vinokurov, L. M. (2003). Expression of the recombinant antibacterial peptide sarcotoxin IA in Escherichia coli cells. *Federation Protein Expression and Purification.*, *28*(2), 350–356. 10465928. Academic Press Inc., Russian. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/3/5/index.htt. https://doi.org/10.1016/S1046-5928(02)00697-6.

Song, X., Wang, J., Wu, F., Li, X., Teng, M., & Gong, W. (2005). cDNA cloning, functional expression and antifungal activities of a dimeric plant defensin SPE10 from Pachyrrhizus erosus seeds. *Plant Molecular Biology*, *57*(1), 13–20. https://doi.org/10.1007/s11103-004-6637-y.

Spirin, A. S. (2004). High-throughput cell-free systems for synthesis of functionally active proteins. *Trends in Biotechnology*, *22*(10), 538–545. 01677799 https://doi.org/10.1016/j.tibtech.2004.08.012.

Srinivasulu, B., Syvitski, R., Seo, J. K., Mattatall, N. R., Knickle, L. C., & Douglas, S. E. (2008). Expression, purification and structural characterization of recombinant hepcidin, an antimicrobial peptide identified in Japanese flounder, Paralichthys olivaceus. *Protein Expression and Purification*, *61*(1), 36–44. 10465928 https://doi.org/10.1016/j.pep.2008.05.012.

Sun, C., Du, X. J., Xu, W. T., Zhang, H. W., Zhao, X. F., & Wang, J. X. (2010). Molecular cloning and characterization of three crustins from the Chinese white shrimp, Fenneropenaeus chinensis. *Fish & Shellfish Immunology*, *28*(4), 517–524. 10959947. Academic Press, China. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/8/3/2/index.htt. https://doi.org/10.1016/j.fsi.2009.12.001.

Sun, Z. Z., Hayes, C. A., Shin, J., Caschera, F., Murray, R. M., & Noireaux, V. (2013). Protocols for implementing an *Escherichia coli* based TX-TL cell-free expression system for synthetic biology. *Journal of Visualized Experiments*, *79*. 1940087X. United States http://www.jove.com/video/50762/protocols-for-implementing-an-escherichia-coli-based-tx-tl-cell-free. https://doi.org/10.3791/50762.

Sun, Y., Li, Q., Li, Z., Zhang, Y., Zhao, J., & Wang, L. (2012). Molecular cloning, expression, purification, and functional characterization of palustrin-2CE, an antimicrobial peptide of Rana chensinensis. *Bioscience, Biotechnology, and Biochemistry*, *76*(1), 157–162. 13476947. China. Available from: http://www.jstage.jst.go.jp/article/bbb/76/1/157/_pdf. https://doi.org/10.1271/bbb.110672.

Sun, X. J., Wang, D. N., Zhang, W. J., & Wu, X. F. (2004). Expression of an antimicrobial peptide identified in the male reproductive system of rats. *Applied Biochemistry and Biotechnology - Part B Molecular Biotechnology*, *28*(3), 185–189. 10736085. Humana Press, China. Available from: http://www.humanapress.com/JournalTOC.pasp?issn=1073-6085. https://doi.org/10.1385/MB:28:3:185.

Supungul, P., Tang, S., Maneeruttanarungroj, C., Rimphanitchayakit, V., Hirono, I., Aoki, T., & Tassanakajon, A. (2008). Cloning, expression and antimicrobial activity of crustinPm1, a major isoform of crustin, from the black tiger shrimp Penaeus monodon. *Developmental and Comparative Immunology*, *32*(1), 61–70. 0145305X https://doi.org/10.1016/j.dci.2007.04.004.

Tacias-Pascacio, V. G., Morellon-Sterling, R., Siar, E. H., Tavano, O., Berenguer-Murcia, Á., & Fernandez-Lafuente, R. (2020). Use of Alcalase in the production of bioactive peptides: A review. *International Journal of Biological Macromolecules*, *165*, 2143–2196. 18790003. Elsevier B.V., Mexico. Available from: http://www.elsevier.com/locate/ijbiomac. https://doi.org/10.1016/j.ijbiomac.2020.10.060.

Takahashi, M. K., Hayes, C. A., Chappell, J., Sun, Z. Z., Murray, R. M., Noireaux, V., & Lucks, J. B. (2015). Characterizing and prototyping genetic networks with cell-free transcription-translation reactions. *Methods*, *86*, 60–72. 10959130. Academic Press Inc., United States. Available from: http://www.elsevier.com/inca/publications/store/6/2/2/9/1/4/index.htt. https://doi.org/10.1016/j.ymeth.2015.05.020.

Tang, X. S., Tang, Z. R., Wang, S. P., Feng, Z. M., Zhou, D., Li, T. J., & Yin, Y. L. (2012). Expression, purification, and antibacterial activity of bovine lactoferrampin-lactoferricin in Pichia pastoris. *Applied Biochemistry and Biotechnology*, *166*(3), 640–651. 15590291 https://doi.org/10.1007/s12010-011-9455-0.

Tavares, L. S., Rettore, J. V., Freitas, R. M., Porto, W. F., Duque, A. P. D. N., Singulani, J. D. L., Silva, O. N., Detoni, M. D. L., Vasconcelos, E. G., Dias, S. C., Franco, O. L., & Santos, M. D. O. (2012). Antimicrobial activity of recombinant Pg-AMP1, a glycine-rich peptide from guava seeds. *Peptides*, *37*(2), 294–300. 18735169 https://doi.org/10.1016/j.peptides.2012.07.017.

Thak, E. J., Yoo, S. J., Moon, H. Y., & Kang, H. A. (2020). Yeast synthetic biology for designed cell factories producing secretory recombinant proteins. *FEMS Yeast Research*, *20*(2). https://doi.org/10.1093/femsyr/foaa009. 1567-1356.

Thomas, J. G., Ayling, A., & Baneyx, F. (1997). Molecular chaperones, folding catalysts, and the recovery of active recombinant proteins from *E. coli*. *Applied Biochemistry and Biotechnology*, *66*(3), 197–238. 0273-2289 https://doi.org/10.1007/bf02785589.

Tian, Z. G., Dong, T. T., Yang, Y. L., Teng, D., & Wang, J. H. (2009). Expression of antimicrobial peptide LH multimers in Escherichia coli C43(DE3). *Applied Microbiology and Biotechnology*, *83*(1), 143–149. 01757598 https://doi.org/10.1007/s00253-009-1893-z.

Tominaga, M., Kondo, A., & Ishii, J. (2022). Engineering of synthetic transcriptional switches in yeast. *Lifestyles*, *12*(4), 557. 2075-1729 https://doi.org/10.3390/life12040557.

Torres-Vanegas, J. D., Cifuentes, J., Puentes, P. R., Quezada, V., Garcia-Brand, A. J., Cruz, J. C., & Reyes, L. H. (2022). Assessing cellular internalization and endosomal escape abilities of novel BUFII-graphene oxide nanobioconjugates. *Frontiers in Chemistry*, *10*. 22962646. Frontiers Media S.A., Colombia. Available from: http://journal.frontiersin.org/journal/chemistry. https://doi.org/10.3389/fchem.2022.974218.

Torres-Vanegas, J. D., Cruz, J. C., & Reyes, L. H. (2021). Delivery systems for nucleic acids and proteins: Barriers, cell capture pathways and nanocarriers. *Pharmaceutics.*, *13*(3). 19994923. MDPI AG, Colombia. Available from: https://www.mdpi.com/1999-4923/13/3/428/pdf. https://doi.org/10.3390/pharmaceutics13030428.

Van Reenen, C. A., Chikindas, M. L., Van Zyl, W. H., & Dicks, L. M. T. (2003). Characterization and heterologous expression of a class IIa bacteriocin, plantaricin 423 from lactobacillus plantarum 423, in Saccharomyces cerevisiae. *International Journal of Food Microbiology*, *81*(1), 29–40. 01681605 https://doi.org/10.1016/S0168-1605(02)00164-2.

Varsaki, A., Murphy, C., Barczynska, A., Jordan, K., & Carroll, C. (2015). The acid adaptive tolerance response in C ampylobacter jejuni induces a global response, as suggested by proteomics and microarrays. *Microbial Biotechnology*, *8*(6), 974–988. 1751-7915 https://doi.org/10.1111/1751-7915.12302.

Vassilevski, A. A., Kozlov, S. A., & Grishin, E. V. (2008). Antimicrobial peptide precursor structures suggest effective production strategies. *Recent Patents on Inflammation and Allergy Drug Discovery*, *2*(1), 58–63. 1872213X. Russian Federation http://www.ingentaconnect.com/content/ben/iad/2008/00000002/00000001/art00005. https://doi.org/10.2174/187221308783399261.

Wang, Y. Q., & Cai, J. Y. (2007). High-level expression of acidic partner-mediated antimicrobial peptide from tandem genes in Escherichia coli. *Applied Biochemistry and Biotechnology*, *141*(2–3), 203–213. 02732289 https://doi.org/10.1007/BF02729062.

Wang, T., Kumru, O. S., Yi, L., Wang, Y. J., Zhang, J., Kim, J. H., Joshi, S. B., Middaugh, C. R., & Volkin, D. B. (2013). Effect of ionic strength and pH on the physical and chemical stability of a monoclonal antibody antigen-binding fragment. *Journal of Pharmaceutical Sciences*, *102*(8), 2520–2537. 15206017. John Wiley and Sons Inc., United States. Available from: http://www.interscience.wiley.com/jpages/0022-3549. https://doi.org/10.1002/jps.23645.

Wang, H., Meng, X. L., Xu, J. P., Wang, J., Wang, H., & Ma, C. W. (2012). Production, purification, and characterization of the cecropin from Plutella xylostella, pxCECA1, using an intein-induced self-cleavable system in Escherichia coli. *Applied Microbiology and Biotechnology*, *94*(4), 1031–1039. 14320614 https://doi.org/10.1007/s00253-011-3863-5.

Wang, A., Su, Y., Wang, S., Shen, M., Chen, F., Chen, M., Ran, X., Cheng, T., & Wang, J. (2010). High efficiency preparation of bioactive human α-defensin 6 in Escherichia coli origami(DE3)pLysS by soluble fusion expression. *Applied Microbiology and Biotechnology*, *87*(5), 1935–1942. 01757598 https://doi.org/10.1007/s00253-010-2688-y.

Wang, A., Wang, S., Shen, M., Chen, F., Zou, Z., Ran, X., Cheng, T., Su, Y., & Wang, J. (2009). High level expression and purification of bioactive human α-defensin 5 mature peptide in Pichia pastoris. *Applied Microbiology and Biotechnology*, *84*(5), 877–884. 01757598 https://doi.org/10.1007/s00253-009-2020-x.

Wang, Q., Zhong, C., & Xiao, H. (2020). Genetic engineering of filamentous Fungi for efficient protein expression and secretion. *Frontiers in Bioengineering and Biotechnology*, *8*. https://doi.org/10.3389/fbioe.2020.00293. 2296-4185.

Wang, Q., Zhu, F., Xin, Y., Liu, J., Luo, L., & Yin, Z. (2011). Expression and purification of antimicrobial peptide buforin IIb in Escherichia coli. *Biotechnology Letters*, *33*(11), 2121–2126. 15736776 https://doi.org/10.1007/s10529-011-0687-4.

Ward, O. P. (2012). Production of recombinant proteins by filamentous fungi. *Biotechnology Advances*, *30*(5), 1119–1139. 07349750 https://doi.org/10.1016/j.biotechadv.2011.09.012.

Wimmerova, L., Keken, Z., Solcova, O., & Vavrova, K. (2023). A comparative analysis of environmental impacts of operational phases of three selected microalgal cultivation systems. *Sustainability.*, *15*(1), 769. 2071-1050 https://doi.org/10.3390/su15010769.

Wong, A., Bryzek, D., Dobosz, E., Scavenius, C., Svoboda, P., Rapala-Kozik, M., Lesner, A., Frydrych, I., Enghild, J., Mydel, P., Pohl, J., Thompson, P. R., Potempa, J., & Koziel, J. (2018). A novel biological role for peptidyl-arginine deiminases: Citrullination of cathelicidin LL-37 controls the Immunostimulatory potential of cell-free DNA. *The Journal of Immunology.*, *200*(7), 2327–2340. 0022-1767 https://doi.org/10.4049/jimmunol.1701391.

Wong, J. F., Hong, H. J., Foo, S. C., Yap, M. K. K., & Tan, J. W. (2022). A review on current and future advancements for commercialized microalgae species. *Food Science and Human Wellness*, *11*(5), 1156–1170. 22134530 https://doi.org/10.1016/j.fshw.2022.04.007.

Wu, G. Q., Li, L. X., Ding, J. X., Wen, L. Z., & Shen, Z. L. (2008). High-level expression and novel purification strategy of recombinant thanatin analog in Escherichia coli. *Current Microbiology*, *57*(2), 95–101. 14320991 https://doi.org/10.1007/s00284-008-9106-z.

Wu, J., Wang, C., He, H., Hu, G., Yang, H., Gao, Y., & Zhong, J. (2011). Molecular analysis and recombinant expression of bovine neutrophil β-defensin 12 and its antimicrobial activity. *Molecular Biology Reports*, *38*(1), 429–436. 15734978 https://doi.org/10.1007/s11033-010-0125-z.

Xia, L, Zhang, F, Liu, Z, Ma, J, & Yang, J. (2013). Expression and characterization of cecropinXJ, a bioactive antimicrobial peptide from *Bombyx mori* (Bombycidae, Lepidoptera) in *Escherichia coli*. *Experimental and Therapeutic Medicine*, *5*(6), 1745–1751. https://doi.org/10.3892/etm.2013.1056. Epub 2013 Apr 9 PMID: 23837066. PMCID: PMC3702707.

Xie, Y. G., Han, F. F., Luan, C., Zhang, H. W., Feng, J., Choi, Y. J., … Wang, Y. Z. (2013a). High-yield soluble expression and simple purification of the antimicrobial peptide OG2 using the intein system in Escherichia coli. *BioMed Research International*, *2013*, 754319. https://doi.org/10.1155/2013/754319. Epub 2013 Jul 2. PMID: 23936842; PMCID: PMC3713655.

Xie, Y. G., Luan, C., Zhang, H. W., Han, F. F., Feng, J., Choi, Y. J., et al. (2013b). Effects of thioredoxin, SUMO and intein on soluble fusion expression of an antimicrobial peptide OG2 in *E. coli*. *Protein & Peptide Letters*, *20*, 54–60.

Xu, M. Q., & Evans, T. C. (2003). Purification of recombinant proteins from *E. coli* by engineered inteins. *Methods in Molecular Biology (Clifton, N.J.)*, *205*, 43–68. 10643745.

Xu, X., Jin, F., Yu, X., Ji, S., Wang, J., Cheng, H., Wang, C., & Zhang, W. (2007). Expression and purification of a recombinant antibacterial peptide, cecropin, from *E. coli*. *Protein Expression and Purification*, *53*(2), 293–301. 10465928 https://doi.org/10.1016/j.pep.2006.12.020.

Xu, X., Jin, F., Yu, X., Ren, S., Hu, J., & Zhang, W. (2007). High-level expression of the recombinant hybrid peptide cecropinA(1-8)-magainin2(1−12) with an ubiquitin fusion partner in *E. coli*. *Protein Expression and Purification*, *55*(1), 175–182. 10465928 https://doi.org/10.1016/j.pep.2007.04.018.

Xu, Z., Peng, L., Zhong, Z., Fang, X., & Cen, P. (2006). High-level expression of a soluble functional antimicrobial peptide, human β-defensin 2, in *E. coli*. *Biotechnology Progress*, *22*(2), 382–386. 87567938 https://doi.org/10.1021/bp0502680.

Xu, Z., Zhong, Z., Huang, L., Peng, L., Wang, F., & Cen, P. (2006). High-level production of bioactive human beta-defensin-4 in *E. coli* by soluble fusion expression. *Applied Microbiology and Biotechnology*, *72*(3), 471–479. 01757598 https://doi.org/10.1016/j.reactfunctpolym.2005.10.008.

Yalçınkaya, D. (2017). *Recombinant human growth hormone production under double promoters by pichia pastoris* (MS thesis).

Yu, R., Dong, S., Zhu, Y., Jin, H., Gao, M., Duan, Z., Zheng, Z., Shi, Z., & Li, Z. (2010). Effective and stable porcine interferon-α production by pichia pastoris fed-batch cultivation with multi-variables clustering and analysis. *Bioprocess and Biosystems Engineering*, *33*(4), 473–483. 16157591 https://doi.org/10.1007/s00449-009-0356-3.

Zahrl, R. J., Prielhofer, R., Ata, Ö., Baumann, K., Mattanovich, D., & Gasser, B. (2022). Pushing and pulling proteins into the yeast secretory pathway enhances recombinant protein secretion. *Metabolic Engineering*, *74*, 36–48. 10967176 https://doi.org/10.1016/j.ymben.2022.08.010.

Zhang, L., Guo, W., & Lu, Y. (2020). Advances in cell-free biosensors: Principle, mechanism, and applications. *Biotechnology Journal*, *15*(9). 18607314. Wiley-VCH Verlag, China. Available from: http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1860-7314. https://doi.org/10.1002/biot.202000187.

Zhang, J., Reddy, J., Buckland, B., & Greasham, R. (2003). Toward consistent and productive complex media for industrial fermentations: Studies on yeast extract for a recombinant yeast fermentation process. *Biotechnology and Bioengineering*, *82*(6), 640–652. https://doi.org/10.1002/bit.10608.

Zhang, M., Shan, Y., Gao, H., Wang, B., Liu, X., Dong, Y., Liu, X., Yao, N., Zhou, Y., Li, X., & Li, H. (2018). Expression of a recombinant hybrid antimicrobial peptide magainin II-cecropin B in the mycelium of the medicinal fungus Cordyceps militaris and its validation in mice. *Microbial Cell Factories*, *17*(1). 14752859. BioMed Central Ltd., China. Available from: http://www.microbialcellfactories.com/home/. https://doi.org/10.1186/s12934-018-0865-3.

Zhang, J., Yang, Y., Teng, D., Tian, Z., Wang, S., & Wang, J. (2011). Expression of plectasin in Pichia pastoris and its characterization as a new antimicrobial peptide against Staphyloccocus and Streptococcus. *Protein Expression and Purification*, *78*(2), 189–196. 10465928 https://doi.org/10.1016/j.pep.2011.04.014.

Zhang, J., Zhang, S. Q., Wu, X., Chen, Y. Q., & Diao, Z. Y. (2006). Expression and characterization of antimicrobial peptide ABP-CM4 in methylotrophic yeast Pichia pastoris. *Process Biochemistry*, *41*(2), 251–256. 13595113 https://doi.org/10.1016/j.procbio.2005.06.030.

Zhao, P., & Cao, G. (2012). Production of bioactive sheep β-defensin-1 in Pichia pastoris. *Journal of Industrial Microbiology and Biotechnology*, ***39***(1), 11–17. 1476-5535 https://doi.org/10.1007/s10295-011-0992-x.

Zhao, M.-X., Chi, Z., Chi, Z.-M., & Madzak, C. (2013). The simultaneous production of single-cell protein and a recombinant antibacterial peptide by expression of an antibacterial peptide gene in Yarrowia lipolytica. *Process Biochemistry*, *48*(2), 212–217. 13595113 https://doi.org/10.1016/j.procbio.2013.01.003.

Zhong, Z., Xu, Z., Peng, L., Huang, L., Fang, X., & Cen, P. (2006). Tandem repeat mhBD2 gene enhance the soluble fusion expression of hBD2 in *E. coli*. *Applied Microbiology and Biotechnology*, *71*(5), 661–667. 01757598 https://doi.org/10.1007/s00253-005-0212-6.

Zhou, Y.-X., Cao, W., Luo, Q.-P., Ma, Y.-S., Wang, J.-Z., & Wei, D.-Z. (2005). Production and purification of a novel antibiotic peptide, adenoregulin, from a recombinant *E. coli*. *Biotechnology Letters*, *27*(10), 725–730. 0141-5492 https://doi.org/10.1007/s10529-005-5361-2.

Zhou, Y., Cao, W., Wang, J., Ma, Y., & Wei, D. (2005). Comparison of expression of monomeric and multimeric adenoregulin genes in *E. coli* and Pichia pastorias. *Protein and Peptide Letters*, *12*(4), 349–355. https://doi.org/10.2174/0929866053765671.

Zhou, L., Lin, Q., Li, B., Li, N., & Zhang, S. (2009). Expression and purification the antimicrobial peptide CM4 in *E. coli*. *Biotechnology Letters*, *31*(3), 437–441. 15736776 https://doi.org/10.1007/s10529-008-9893-0.

Zhou, Q. F., Luo, X. G., Ye, L., & Xi, T. (2007). High-level production of a novel antimicrobial peptide perinerin in *E. coli* by fusion expression. *Current Microbiology*, *54*(5), 366–370. 14320991 https://doi.org/10.1007/s00284-006-0466-y.

Zhou, L., Zhao, Z., Li, B., Cai, Y., & Zhang, S. (2009). TrxA mediating fusion expression of antimicrobial peptide CM4 from multiple joined genes in *E. coli*. *Protein Expression and Purification*, *64*(2), 225–230. 10465928 https://doi.org/10.1016/j.pep.2008.11.006.

Zorko, M., & Jerala, R. (2010). Production of recombinant antimicrobial peptides in bacteria. *Methods in Molecular Biology (Clifton, N.J.)*, *618*, 61–76. 19406029 https://doi.org/10.1007/978-1-60,761-594-1_5.

ANTIMICROBIAL PEPTIDES

A Roadmap for Accelerating Discovery and Development

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Contains a comprehensive description of computational and experimental approaches for the discovery, manufacture, and clinical translation of antimicrobial peptides.

Antimicrobial Peptides: A Roadmap for Accelerating Discovery and Development covers the most important efforts of scientists and engineers worldwide to accelerate the process of discovery, production, and eventual market penetration of more potent antimicrobial peptides. These efforts have been fueled by emerging technologies such as artificial intelligence and data science, molecular and CFD simulations, easy-to-use process simulation packages, microfluidics, and 3D printing, among many others. Such technologies can now be implemented and scaled up quickly and at relatively low cost in low-budget production facilities, critical to moving to sustainable and marketable products worldwide.

Discovering novel antimicrobial peptides rationally and cost-effectively has emerged as one of the significant challenges of modern biotechnology. Thus far, this process has been tedious and costly, resulting in molecules with activities far below those needed to address the current challenge of microbial resistance to antibiotics that takes the lives of thousands of people around the world every year. Due to these issues, large biopharmaceutical companies have decided not to invest in discovering novel antimicrobials, which is worrisome in the long run due to the possibility of a new pandemic.

This book also delineates how multidisciplinary teams have assembled to address the challenges of manufacturing, biological testing, and clinical trials to finally reach complete translation. It is an ideal resource for scientists, researchers, engineers, and students in the fields of biotechnology, microbiology, biochemistry, and pharmaceuticals. Finally, anyone interested in the emerging technologies that are enabling the discovery and development of new antimicrobial peptides may find this book interesting and informative.

Key Features

- Covers computational tools (including emerging artificial intelligence algorithms) and microfluidic systems for discovering and high-throughput screening of AMPs
- Discusses the application of bioprocess engineering scale-up approaches for AMPs production and purification with the aid of process simulation tools and rapid prototyping
- Highlights user-centered design and formulation of products with AMPs
- Describes the whole pipeline for AMPs production

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